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Lee

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(54) **BUILDING MANAGEMENT SYSTEMS AND METHODS FOR TUNING FAULT DETECTION THRESHOLDS**

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(56) **References Cited**

U.S. PATENT DOCUMENTS

5,301,109 A 4/1994 Landauer et al.
5,446,677 A 8/1995 Jensen et al.
5,581,478 A 12/1996 Cruse et al.
(Continued)

FOREIGN PATENT DOCUMENTS

AU 2019226217 A1 11/2020
AU 2019226264 A1 11/2020
(Continued)

OTHER PUBLICATIONS

Cronrath et al., "Enhancing Digital Twins through Reinforcement Learning," 2019 IEEE 15th International Conference on Automation Science and Engineering (CASE), 2019 (6 pages).

(Continued)

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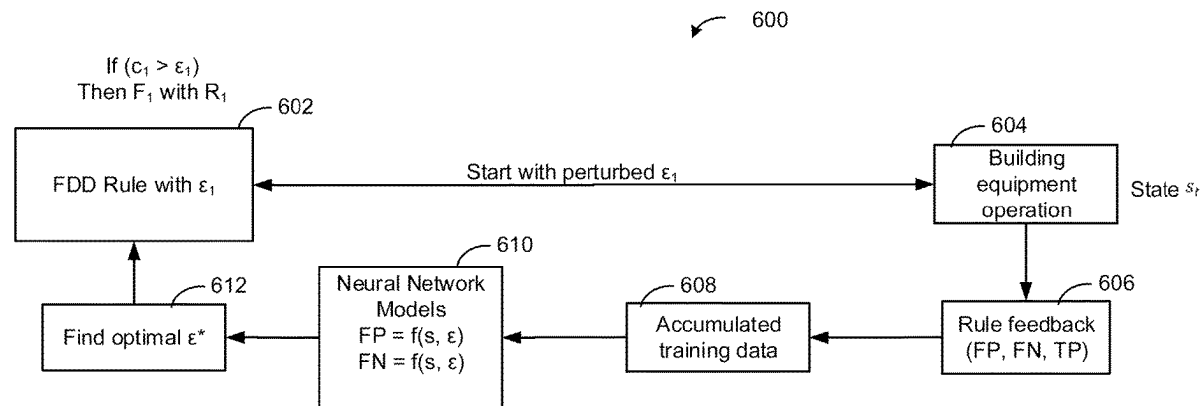
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ABSTRACT

Building management systems and methods for autonomously tuning rule thresholds are disclosed. In one aspect, the method includes a providing a rule including a threshold, the rule used to determine whether building equipment has a fault. The method further includes receiving a state of the building equipment, assessing using a machine learning model whether the determination of whether the building equipment has a fault is a false positive or a false negative based on the state and the threshold, determining a new threshold based on the assessment of the machine learning model, and replacing the threshold with the new threshold to make subsequent determinations of whether the building equipment or other building equipment has a fault.

18 Claims, 8 Drawing Sheets



(56)

References Cited

U.S. PATENT DOCUMENTS

5,812,962	A	9/1998	Kovac	9,838,844	B2	12/2017	Emeis et al.
5,960,381	A	9/1999	Singers et al.	9,886,478	B2	2/2018	Mukherjee
5,973,662	A	10/1999	Singers et al.	9,948,359	B2	4/2018	Horton
6,014,612	A	1/2000	Larson et al.	10,055,114	B2	8/2018	Shah et al.
6,031,547	A	2/2000	Kennedy	10,055,206	B2	8/2018	Park et al.
6,134,511	A	10/2000	Subbarao	10,116,461	B2	10/2018	Fairweather et al.
6,157,943	A	12/2000	Meyer	10,169,454	B2	1/2019	Ait-Mokhtar et al.
6,285,966	B1	9/2001	Brown et al.	10,171,297	B2	1/2019	Stewart et al.
6,363,422	B1	3/2002	Hunter et al.	10,171,586	B2	1/2019	Shaashua et al.
6,385,510	B1	5/2002	Hoog et al.	10,187,258	B2	1/2019	Nagesh et al.
6,389,331	B1	5/2002	Jensen et al.	10,514,963	B2	12/2019	Shrivastava et al.
6,401,027	B1	6/2002	Xu et al.	10,515,098	B2	12/2019	Park et al.
6,437,691	B1	8/2002	Sandelman et al.	10,521,968	B2	12/2019	Locke et al.
6,477,518	B1	11/2002	Li et al.	10,534,326	B2	1/2020	Sridharan et al.
6,487,457	B1	11/2002	Hull et al.	10,536,295	B2	1/2020	Fairweather et al.
6,493,755	B1	12/2002	Hansen et al.	10,564,993	B2	2/2020	Deutsch et al.
6,577,323	B1	6/2003	Jamieson et al.	10,705,492	B2	7/2020	Harvey
6,626,366	B2	9/2003	Kayahara et al.	10,708,078	B2	7/2020	Harvey
6,646,660	B1	11/2003	Patty	10,739,029	B2	8/2020	Sinha et al.
6,704,016	B1	3/2004	Oliver et al.	10,747,183	B2	8/2020	Sinha et al.
6,732,540	B2	5/2004	Sugihara et al.	10,760,815	B2	9/2020	Janakiraman et al.
6,764,019	B1	7/2004	Kayahara et al.	10,762,475	B2	9/2020	Song et al.
6,782,385	B2	8/2004	Natsumeda et al.	10,824,120	B2	11/2020	Ahmed
6,813,532	B2	11/2004	Eryurek et al.	10,845,771	B2	11/2020	Harvey
6,816,811	B2	11/2004	Seem	10,854,194	B2	12/2020	Park et al.
6,823,680	B2	11/2004	Jayanth	10,862,928	B1	12/2020	Badawy et al.
6,826,454	B2	11/2004	Sulfstede	10,901,373	B2	1/2021	Locke et al.
6,865,511	B2	3/2005	Frerichs et al.	10,921,760	B2	2/2021	Harvey
6,925,338	B2	8/2005	Eryurek et al.	10,921,972	B2	2/2021	Park et al.
6,986,138	B1	1/2006	Sakaguchi et al.	10,969,133	B2	4/2021	Harvey
7,031,880	B1	4/2006	Seem et al.	10,986,121	B2	4/2021	Stockdale et al.
7,401,057	B2	7/2008	Eder	11,016,998	B2	5/2021	Park et al.
7,552,467	B2	6/2009	Lindsay	11,024,292	B2	6/2021	Park et al.
7,627,544	B2	12/2009	Chkodrov et al.	11,038,709	B2	6/2021	Park et al.
7,818,249	B2	10/2010	Lovejoy et al.	11,041,650	B2	6/2021	Li et al.
7,889,051	B1	2/2011	Billig et al.	11,054,796	B2	7/2021	Holaso
7,996,488	B1	8/2011	Casabella et al.	11,070,390	B2	7/2021	Park et al.
8,078,330	B2	12/2011	Brickfield et al.	11,073,976	B2	7/2021	Park et al.
8,104,044	B1	1/2012	Scofield et al.	11,085,663	B2	8/2021	Ellis et al.
8,229,470	B1	7/2012	Ranjan et al.	11,108,587	B2	8/2021	Park et al.
8,401,991	B2	3/2013	Wu et al.	11,113,295	B2	9/2021	Park et al.
8,495,745	B1	7/2013	Schrecker et al.	11,120,012	B2	9/2021	Park et al.
8,516,016	B2	8/2013	Park et al.	11,131,473	B2	9/2021	Risbeck et al.
8,532,808	B2	9/2013	Drees et al.	11,210,591	B2	12/2021	Alanqar et al.
8,532,839	B2	9/2013	Drees et al.	11,215,375	B2	1/2022	Alanqar et al.
8,600,556	B2	12/2013	Nesler et al.	11,229,138	B1	1/2022	Harvey et al.
8,635,182	B2	1/2014	Mackay	11,243,503	B2	2/2022	Przybylski et al.
8,682,921	B2	3/2014	Park et al.	11,274,849	B2	3/2022	Bell et al.
8,731,724	B2	5/2014	Drees et al.	11,276,125	B2	3/2022	Pancholi et al.
8,737,334	B2	5/2014	Ahn et al.	11,314,726	B2	4/2022	Park et al.
8,738,334	B2	5/2014	Jiang et al.	11,314,788	B2	4/2022	Park et al.
8,751,487	B2	6/2014	Byrne et al.	11,556,105	B2	1/2023	Cooley et al.
8,788,097	B2	7/2014	Drees et al.	11,561,522	B2	1/2023	Cooley et al.
8,805,995	B1	8/2014	Oliver	11,561,523	B2	1/2023	Cooley et al.
8,843,238	B2	9/2014	Wenzel et al.	11,573,551	B2	2/2023	Cooley et al.
8,874,071	B2	10/2014	Sherman et al.	11,586,167	B2	2/2023	Cooley et al.
8,941,465	B2	1/2015	Pineau et al.	2002/0010562	A1	1/2002	Schleiss et al.
8,990,127	B2	3/2015	Taylor	2002/0016639	A1	2/2002	Smith et al.
9,070,113	B2	6/2015	Shafiee et al.	2002/0059229	A1	5/2002	Natsumeda et al.
9,116,978	B2	8/2015	Park et al.	2002/0123864	A1	9/2002	Eryurek et al.
9,185,095	B1	11/2015	Moritz et al.	2002/0147506	A1	10/2002	Eryurek et al.
9,189,527	B2	11/2015	Park et al.	2002/0177909	A1	11/2002	Fu et al.
9,196,009	B2	11/2015	Drees et al.	2003/0005486	A1	1/2003	Ridolfo et al.
9,229,966	B2	1/2016	Aymeloglu et al.	2003/0014130	A1	1/2003	Grumelart
9,286,582	B2	3/2016	Drees et al.	2003/0073432	A1	4/2003	Meade, II
9,311,807	B2	4/2016	Schultz et al.	2003/0158704	A1	8/2003	Triginai et al.
9,344,751	B1	5/2016	Ream et al.	2003/0171851	A1	9/2003	Brickfield et al.
9,354,968	B2	5/2016	Wenzel et al.	2003/0200059	A1	10/2003	Ignatowski et al.
9,507,686	B2	11/2016	Horn et al.	2004/0068390	A1	4/2004	Saunders
9,524,594	B2	12/2016	Ouyang et al.	2004/0128314	A1	7/2004	Katibah et al.
9,558,196	B2	1/2017	Johnston et al.	2004/0133314	A1	7/2004	Ehlers et al.
9,652,813	B2	5/2017	Gifford et al.	2004/0199360	A1	10/2004	Friman et al.
9,753,455	B2	9/2017	Drees	2005/0055308	A1	3/2005	Meyer et al.
9,811,249	B2	11/2017	Chen et al.	2005/0108262	A1	5/2005	Fawcett et al.
9,817,383	B1	11/2017	Sinha et al.	2005/0154494	A1	7/2005	Ahmed
				2005/0278703	A1	12/2005	Lo et al.
				2005/0283337	A1	12/2005	Sayal
				2006/0095521	A1	5/2006	Patinkin
				2006/0140207	A1	6/2006	Eschbach et al.

(56)	References Cited			2014/0032506	A1	1/2014	Hoey et al.	
	U.S. PATENT DOCUMENTS			2014/0059483	A1	2/2014	Mairs et al.	
				2014/0081652	A1	3/2014	Klindworth	
				2014/0107993	A1	4/2014	Cheng	
2006/0184479	A1	8/2006	Levine	2014/0135952	A1	5/2014	Maehara	
2006/0200476	A1	9/2006	Gottumukkala et al.	2014/0152651	A1	6/2014	Chen et al.	
2006/0265751	A1	11/2006	Cosquer et al.	2014/0172184	A1	6/2014	Schmidt et al.	
2006/0271589	A1	11/2006	Horowitz et al.	2014/0189861	A1	7/2014	Gupta et al.	
2007/0028179	A1	2/2007	Levin et al.	2014/0207282	A1	7/2014	Angle et al.	
2007/0203693	A1	8/2007	Estes	2014/0258052	A1	9/2014	Khuti et al.	
2007/0261062	A1	11/2007	Bansal et al.	2014/0269614	A1	9/2014	Maguire et al.	
2007/0273497	A1	11/2007	Kuroda et al.	2014/0277765	A1	9/2014	Karimi et al.	
2007/0273610	A1	11/2007	Baillot	2014/0278461	A1	9/2014	Artz	
2008/0034425	A1	2/2008	Overcash et al.	2014/0327555	A1	11/2014	Sager et al.	
2008/0094230	A1	4/2008	Mock et al.	2015/0019174	A1	1/2015	Kiff et al.	
2008/0097816	A1	4/2008	Freire et al.	2015/0042240	A1	2/2015	Aggarwal et al.	
2008/0186160	A1	8/2008	Kim et al.	2015/0105917	A1	4/2015	Sasaki et al.	
2008/0249756	A1	10/2008	Chaisuparasmikul	2015/0145468	A1	5/2015	Ma et al.	
2008/0252723	A1	10/2008	Park	2015/0156031	A1	6/2015	Fadell et al.	
2008/0281472	A1	11/2008	Podgorny et al.	2015/0168931	A1	6/2015	Jin	
2009/0195349	A1	8/2009	Frader-Thompson et al.	2015/0172300	A1	6/2015	Cochenour	
2009/0271169	A1	10/2009	Minto et al.	2015/0178421	A1	6/2015	Borrelli et al.	
2010/0045439	A1	2/2010	Tak et al.	2015/0185261	A1	7/2015	Frader-Thompson et al.	
2010/0058248	A1	3/2010	Park	2015/0186777	A1	7/2015	Lecue et al.	
2010/0131533	A1	5/2010	Ortiz	2015/0202962	A1	7/2015	Habashima et al.	
2010/0274366	A1	10/2010	Fata et al.	2015/0204563	A1	7/2015	Imes et al.	
2010/0281387	A1	11/2010	Holland et al.	2015/0235267	A1	8/2015	Steube et al.	
2010/0286937	A1	11/2010	Hedley et al.	2015/0241895	A1	8/2015	Lu et al.	
2010/0324962	A1	12/2010	Nesler et al.	2015/0244730	A1	8/2015	Vu et al.	
2011/0015802	A1	1/2011	Imes	2015/0244732	A1	8/2015	Golshan et al.	
2011/0047418	A1	2/2011	Drees et al.	2015/0261863	A1	9/2015	Dey et al.	
2011/0061015	A1	3/2011	Drees et al.	2015/0263900	A1	9/2015	Polyakov et al.	
2011/0071685	A1	3/2011	Huneycutt et al.	2015/0286969	A1	10/2015	Warner et al.	
2011/0077950	A1	3/2011	Hughston	2015/0295796	A1	10/2015	Hsiao et al.	
2011/0087650	A1	4/2011	Mackay et al.	2015/0304193	A1	10/2015	Ishii et al.	
2011/0087988	A1	4/2011	Ray et al.	2015/0316918	A1	11/2015	Schleiss et al.	
2011/0088000	A1	4/2011	Mackay	2015/0324422	A1	11/2015	Elder	
2011/0125737	A1	5/2011	Pothering et al.	2015/0341212	A1	11/2015	Hsiao et al.	
2011/0137853	A1	6/2011	Mackay	2015/0348417	A1	12/2015	Ignaczak et al.	
2011/0153603	A1	6/2011	Adiba et al.	2015/0379080	A1	12/2015	Jochimski	
2011/0154363	A1	6/2011	Karmarkar	2015/0379429	A1*	12/2015	Lee	G09B 5/00 706/11
2011/0157357	A1	6/2011	Weisensale et al.					
2011/0178977	A1	7/2011	Drees	2016/0011753	A1	1/2016	McFarland et al.	
2011/0191343	A1	8/2011	Heaton et al.	2016/0033946	A1	2/2016	Zhu et al.	
2011/0205022	A1	8/2011	Cavallaro et al.	2016/0035246	A1	2/2016	Curtis	
2011/0218777	A1	9/2011	Chen et al.	2016/0065601	A1	3/2016	Gong et al.	
2012/0011126	A1	1/2012	Park et al.	2016/0070736	A1	3/2016	Swan et al.	
2012/0011141	A1	1/2012	Park et al.	2016/0078229	A1	3/2016	Gong et al.	
2012/0022698	A1	1/2012	Mackay	2016/0090839	A1	3/2016	Stolarczyk	
2012/0062577	A1	3/2012	Nixon	2016/0119434	A1	4/2016	Dong et al.	
2012/0064923	A1	3/2012	Imes et al.	2016/0127712	A1	5/2016	Alfredsson et al.	
2012/0083930	A1	4/2012	Ilic et al.	2016/0139752	A1	5/2016	Shim et al.	
2012/0100825	A1	4/2012	Sherman et al.	2016/0163186	A1	6/2016	Davidson et al.	
2012/0101637	A1	4/2012	Imes et al.	2016/0170390	A1	6/2016	Xie et al.	
2012/0135759	A1	5/2012	Imes et al.	2016/0171862	A1	6/2016	Das et al.	
2012/0136485	A1	5/2012	Weber et al.	2016/0173816	A1	6/2016	Huenerfauth et al.	
2012/0158633	A1	6/2012	Eder	2016/0179315	A1	6/2016	Sarao et al.	
2012/0259583	A1	10/2012	Noboa et al.	2016/0179342	A1	6/2016	Sarao et al.	
2012/0272228	A1	10/2012	Marndi et al.	2016/0179990	A1	6/2016	Sarao et al.	
2012/0278051	A1	11/2012	Jiang et al.	2016/0195856	A1	7/2016	Spero	
2013/0007063	A1	1/2013	Kalra et al.	2016/0203036	A1*	7/2016	Mezic	G06F 11/0751 714/819
2013/0038430	A1	2/2013	Blower et al.					
2013/0038707	A1	2/2013	Cunningham et al.	2016/0212165	A1	7/2016	Singla et al.	
2013/0060820	A1	3/2013	Bulusu et al.	2016/0239660	A1	8/2016	Azvine et al.	
2013/0086497	A1	4/2013	Ambuhl et al.	2016/0239756	A1	8/2016	Aggour et al.	
2013/0097706	A1	4/2013	Titonis et al.	2016/0247129	A1	8/2016	Song et al.	
2013/0103221	A1	4/2013	Raman et al.	2016/0260063	A1	9/2016	Harris et al.	
2013/0167035	A1	6/2013	Imes et al.	2016/0313751	A1	10/2016	Risbeck et al.	
2013/0170710	A1	7/2013	Kuoch et al.	2016/0313752	A1	10/2016	Przybylski	
2013/0204836	A1	8/2013	Choi et al.	2016/0313902	A1	10/2016	Hill et al.	
2013/0246916	A1	9/2013	Reimann et al.	2016/0350364	A1	12/2016	Anicic et al.	
2013/0247205	A1	9/2013	Schrecker et al.	2016/0357521	A1	12/2016	Zhang et al.	
2013/0262035	A1	10/2013	Mills	2016/0357828	A1	12/2016	Tobin et al.	
2013/0275174	A1	10/2013	Bennett et al.	2016/0358432	A1	12/2016	Branscomb et al.	
2013/0275908	A1	10/2013	Reichard	2016/0363336	A1	12/2016	Roth et al.	
2013/0297050	A1	11/2013	Reichard et al.	2016/0370258	A1	12/2016	Perez	
2013/0298244	A1	11/2013	Kumar et al.	2016/0378306	A1	12/2016	Kresl et al.	
2013/0331995	A1	12/2013	Rosen	2016/0379326	A1	12/2016	Chan-Gove et al.	
2013/0338970	A1	12/2013	Reggetti	2017/0006135	A1	1/2017	Siebel	

(56)

References Cited

U.S. PATENT DOCUMENTS

2017/0011318	A1	1/2017	Vigano et al.	2019/0121801	A1	4/2019	Jethwa et al.
2017/0017221	A1	1/2017	Lamparter et al.	2019/0138512	A1	5/2019	Pourmohammad et al.
2017/0039255	A1	2/2017	Raj et al.	2019/0139315	A1	5/2019	Locke et al.
2017/0052536	A1	2/2017	Warner et al.	2019/0147883	A1	5/2019	Mellenthin et al.
2017/0053441	A1	2/2017	Nadumane et al.	2019/0158309	A1	5/2019	Park et al.
2017/0063894	A1	3/2017	Muddu et al.	2019/0163152	A1	5/2019	Worrall et al.
2017/0068409	A1	3/2017	Nair	2019/0268178	A1	8/2019	Fairweather et al.
2017/0070775	A1	3/2017	Taxier et al.	2019/0310979	A1	10/2019	Masuzaki et al.
2017/0075984	A1	3/2017	Deshpande et al.	2019/0361411	A1	11/2019	Park et al.
2017/0084168	A1	3/2017	Janchookiat	2019/0361412	A1	11/2019	Park et al.
2017/0090437	A1	3/2017	Veeramani et al.	2019/0377306	A1	12/2019	Harvey
2017/0093700	A1	3/2017	Gilley et al.	2020/0076196	A1	3/2020	Lee et al.
2017/0098086	A1	4/2017	Hoerneck et al.	2020/0092127	A1	3/2020	Park et al.
2017/0103327	A1	4/2017	Penilla et al.	2020/0133213	A1	4/2020	Park et al.
2017/0103403	A1	4/2017	Chu et al.	2020/0159182	A1	5/2020	Goyal
2017/0123389	A1	5/2017	Baez et al.	2020/0226156	A1	7/2020	Borra et al.
2017/0134415	A1	5/2017	Muddu et al.	2020/0285203	A1	9/2020	Thakur et al.
2017/0177715	A1	6/2017	Chang et al.	2020/0313924	A1	10/2020	Park et al.
2017/0180147	A1	6/2017	Brandman et al.	2020/0336328	A1	10/2020	Harvey
2017/0188216	A1	6/2017	Koskas et al.	2020/0348632	A1	11/2020	Harvey
2017/0212482	A1	7/2017	Boettcher et al.	2020/0358794	A1*	11/2020	Vasseur G06F 16/285
2017/0212668	A1	7/2017	Shah et al.	2020/0370771	A1	11/2020	Alanqar et al.
2017/0220641	A1	8/2017	Chi et al.	2020/0387576	A1	12/2020	Brett et al.
2017/0230930	A1	8/2017	Frey	2020/0396208	A1	12/2020	Brett et al.
2017/0235817	A1	8/2017	Deodhar et al.	2021/0010351	A1	1/2021	Sun et al.
2017/0251182	A1	8/2017	Siminoff et al.	2021/0025612	A1	1/2021	Sinha et al.
2017/0270124	A1	9/2017	Nagano et al.	2021/0042299	A1	2/2021	Migliori
2017/0277769	A1	9/2017	Pasupathy et al.	2021/0043221	A1	2/2021	Yelchuru et al.
2017/0278003	A1	9/2017	Liu	2021/0166548	A1*	6/2021	Ellam G08B 21/182
2017/0294132	A1	10/2017	Colmenares	2021/0182859	A1*	6/2021	Srinivasa Rao ... G06Q 20/4016
2017/0315522	A1	11/2017	Kwon et al.	2021/0191826	A1	6/2021	Duraisingh et al.
2017/0315697	A1	11/2017	Jacobson et al.	2021/0208546	A1	7/2021	Locke et al.
2017/0322534	A1	11/2017	Sinha et al.	2021/0240147	A1	8/2021	Papadopoulos et al.
2017/0323389	A1	11/2017	Vavrasek	2021/0258371	A1	8/2021	Sundaresan
2017/0329289	A1	11/2017	Kohn et al.	2021/0325070	A1	10/2021	Endel et al.
2017/0336770	A1	11/2017	MacMillan	2021/0342961	A1	11/2021	Winter et al.
2017/0345287	A1	11/2017	Fuller et al.	2021/0344695	A1*	11/2021	Palani G06N 3/043
2017/0351957	A1	12/2017	Lecue et al.	2021/0381711	A1	12/2021	Harvey et al.
2017/0357225	A1	12/2017	Asp et al.	2021/0381712	A1	12/2021	Harvey et al.
2017/0357490	A1	12/2017	Park et al.	2021/0382445	A1	12/2021	Harvey et al.
2017/0357908	A1	12/2017	Cabadi et al.	2021/0383041	A1	12/2021	Harvey et al.
2018/0012159	A1	1/2018	Kozloski et al.	2021/0383042	A1	12/2021	Harvey et al.
2018/0013579	A1	1/2018	Fairweather et al.	2021/0383200	A1	12/2021	Harvey et al.
2018/0024520	A1	1/2018	Sinha et al.	2021/0383219	A1	12/2021	Harvey et al.
2018/0039238	A1	2/2018	Gärtner et al.	2021/0383235	A1	12/2021	Harvey et al.
2018/0048485	A1	2/2018	Pelton et al.	2021/0383236	A1	12/2021	Harvey et al.
2018/0069932	A1	3/2018	Tiwari et al.	2022/0066402	A1	3/2022	Harvey et al.
2018/0114140	A1	4/2018	Chen et al.	2022/0066405	A1	3/2022	Harvey
2018/0137288	A1	5/2018	Polyakov	2022/0066432	A1	3/2022	Harvey et al.
2018/0157930	A1	6/2018	Rutschman et al.	2022/0066434	A1	3/2022	Harvey et al.
2018/0162400	A1	6/2018	Abdar	2022/0066528	A1	3/2022	Harvey et al.
2018/0176241	A1	6/2018	Manadhata et al.	2022/0066722	A1	3/2022	Harvey et al.
2018/0198627	A1	7/2018	Mullins	2022/0066754	A1	3/2022	Harvey et al.
2018/0203961	A1	7/2018	Aisu et al.	2022/0066761	A1	3/2022	Harvey et al.
2018/0239982	A1	8/2018	Rutschman et al.	2022/0067226	A1	3/2022	Harvey et al.
2018/0275625	A1	9/2018	Park et al.	2022/0067227	A1	3/2022	Harvey et al.
2018/0276962	A1	9/2018	Butler et al.	2022/0067230	A1	3/2022	Harvey et al.
2018/0292797	A1	10/2018	Lamparter et al.	2022/0069863	A1	3/2022	Harvey et al.
2018/0313561	A1	11/2018	Sinha et al.	2022/0070293	A1	3/2022	Harvey et al.
2018/0336785	A1	11/2018	Ghannam et al.	2022/0121965	A1	4/2022	Chatterji et al.
2018/0356775	A1	12/2018	Harvey	2022/0138684	A1	5/2022	Harvey
2018/0359111	A1	12/2018	Harvey	2022/0147000	A1	5/2022	Cooley et al.
2018/0364654	A1	12/2018	Locke et al.	2022/0150124	A1	5/2022	Cooley et al.
2019/0005025	A1	1/2019	Malabarba	2022/0156154	A1*	5/2022	Varnavas G06F 11/3409
2019/0013023	A1	1/2019	Pourmohammad et al.	2022/0215264	A1	7/2022	Harvey et al.
2019/0017719	A1	1/2019	Sinha et al.	2023/0010757	A1	1/2023	Preciado
2019/0025771	A1	1/2019	Park et al.	2023/0071312	A1	3/2023	Preciado et al.
2019/0026359	A1	1/2019	Park et al.	2023/0076011	A1	3/2023	Preciado et al.
2019/0037135	A1	1/2019	Hedge	2023/0083703	A1	3/2023	Meiners
2019/0042988	A1	2/2019	Brown et al.	2023/0214555	A1	7/2023	Harvey et al.
2019/0088106	A1	3/2019	Grundstrom	2023/0252205	A1	8/2023	Harvey et al.
2019/0094824	A1	3/2019	Xie et al.	2023/0334200	A1	10/2023	Meiners
2019/0096217	A1	3/2019	Pourmohammad et al.				
2019/0098113	A1	3/2019	Park et al.				
2019/0102840	A1	4/2019	Perl et al.				

FOREIGN PATENT DOCUMENTS

CN	101415011	A	4/2009
CN	102136099	A	7/2011
CN	102136100	A	7/2011
CN	102650876	A	8/2012

(56)

References Cited

FOREIGN PATENT DOCUMENTS

CN	104040583	A	9/2014
CN	104603832	A	5/2015
CN	104919484	A	9/2015
CN	106204392	A	12/2016
CN	106406806	A	2/2017
CN	106960269	A	7/2017
CN	107147639	A1	9/2017
CN	107598928	A	1/2018
EP	2 528 033	A1	11/2012
EP	3 268 821	B1	1/2018
EP	3 324 306	A1	5/2018
EP	4 226 263	A1	8/2023
JP	H10-049552	A	2/1998
JP	3279078	B2 *	4/2002
JP	2003-162573	A	6/2003
JP	2007-018322	A	1/2007
JP	4073946	B1	4/2008
JP	2008-107930	A	5/2008
JP	2013-152618	A	8/2013
JP	2014-044457	A	3/2014
KR	2016/0102923	A	8/2016
WO	WO-2009/020158	A1	2/2009
WO	WO-2011/100255	A2	8/2011
WO	WO-2013/050333	A1	4/2013
WO	WO-2015/106702	A1	7/2015
WO	WO-2015/145648	A1	10/2015
WO	WO-2017/035536	A1	3/2017
WO	WO-2017/192422	A1	11/2017
WO	WO-2017/192770	A1	11/2017
WO	WO-2017/194244	A1	11/2017
WO	WO-2017/205330	A1	11/2017
WO	WO-2017/213918	A1	12/2017
WO	WO-2018/132112	A1	7/2018
WO	WO-2018/204384	A1	11/2018
WO	WO-2018/232147	A1	12/2018
WO	WO-2019/018304	A1	1/2019
WO	WO-2020/061621	A1	4/2020
WO	WO-2021/163287	A1	8/2021
WO	WO-2022/042925	A1	3/2022
WO	WO-2022/103812	A1	5/2022
WO	WO-2022/103813	A1	5/2022
WO	WO-2022/103820	A1	5/2022
WO	WO-2022/103822	A1	5/2022
WO	WO-2022/103824	A1	5/2022
WO	WO-2022/103829	A1	5/2022
WO	WO-2022/103831	A1	5/2022

OTHER PUBLICATIONS

Ding et al., "MB2C Model-Based Deep Reinforcement Learning for Multi-zone Building Control," BuildSys '20, Virtual Event, Japan, Nov. 18-20, 2020 (pp. 50-59).

Gilles, P., "Graph Models for Systems-of-Systems Digital Twins," Orange Innovation, IT & Services, Meylan, FR, Jun. 2021 (8 pages).

International Search Report and Written Opinion on PCT No. PCT/US2022/050079 dated Mar. 28, 2023 (21 pages).

Balaji et al., "Brick: Metadata schema for portable smart building applications," Applied Energy, 2018 (20 pages).

Balaji et al., "Brick: Metadata schema for portable smart building applications," Applied Energy, Sep. 15, 2018, 3 pages, (Abstract).

Balaji et al., "Demo Abstract: Portable Queries Using the Brick Schema for Building Applications," BuildSys '16, Palo Alto, CA, USA, Nov. 16-17, 2016 (2 pages).

Balaji, B. et al., "Brick: Towards a Unified Metadata Schema For Buildings," BuildSys '16, Palo Alto, CA, USA, Nov. 16-17, 2016 (10 pages).

Bhattacharya et al., "Short Paper: Analyzing Metadata Schemas for Buildings—The Good, The Bad and The Ugly," BuildSys '15, Seoul, South Korea, Nov. 4-5, 2015 (4 pages).

Bhattacharya, A., "Enabling Scalable Smart-Building Analytics," Electrical Engineering and Computer Sciences, University of California at Berkeley, Technical Report No. UCB/EECS-2016-201, Dec. 15, 2016 (121 pages).

Brick, "Brick Schema: Building Blocks for Smart Buildings," URL: chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://www.memoori.com/wp-content/uploads/2016/06/Brick_Schema_Whitepaper.pdf, Mar. 2019 (17 pages).

Brick, "Brick: Towards a Unified Metadata Schema For Buildings," URL: chrome-extension://efaidnbmnnnibpcajpcglclefindmkaj/https://brickschema.org/papers/Brick_BuildSys_Presentation.pdf, Presented at BuildSys '16, Nov. 2016 (46 pages).

Brick, "Metadata Schema for Buildings," URL: https://brickschema.org/docs/Brick-Leaflet.pdf, retrieved from internet Dec. 24, 2019 (3 pages).

Chinese Office Action on CN Appl. No. 201780003995.9 dated Apr. 8, 2021 (21 pages with English language translation).

Chinese Office action on CN Appl. No. 201780043400.2 dated Apr. 25, 2021 (15 pages with English language translation).

Curry, E. et al., "Linking building data in the cloud: Integrating cross-domain building data using linked data," Advanced Engineering Informatics, 2013, 27 (pp. 206-219).

Digital Platform Litigation Documents Part 1, includes cover letter, dismissal of case DDE-1-21-cv-01796, IPR2023-00022 (documents filed Jan. 23, 2023-Oct. 7, 2022), and IPR2023-00085 (documents filed Jan. 26, 2023-Oct. 20, 2022) (748 pages total).

Digital Platform Litigation Documents Part 10, includes DDE-1-21-cv-01796 (documents filed Nov. 1, 2022-Dec. 22, 2021 (1795 pages total).

Digital Platform Litigation Documents Part 2, includes IPR2023-00085 (documents filed Oct. 20, 2022) (172 pages total).

Digital Platform Litigation Documents Part 3, includes IPR2023-00085 (documents filed Oct. 20, 2022) and IPR2023-00170 (documents filed Nov. 28, 2022-Nov. 7, 2022) (397 pages total).

Digital Platform Litigation Documents Part 4, includes IPR2023-00170 (documents filed Nov. 7, 2022) and IPR2023-00217 (documents filed Jan. 18, 2023-Nov. 15, 2022) (434 pages total).

Digital Platform Litigation Documents Part 5, includes IPR2023-00217 (documents filed Nov. 15, 2022) and IPR2023-00257 (documents filed Jan. 25, 2023-Nov. 23, 2022) (316 pages total).

Digital Platform Litigation Documents Part 6, includes IPR2023-00257 (documents filed Nov. 23, 2022) and IPR 2023-00346 (documents filed Jan. 3, 2023-Dec. 13, 2022) (295 pages total).

Digital Platform Litigation Documents Part 7, includes IPR 2023-00346 (documents filed Dec. 13, 2022) and IPR2023-00347 (documents filed Jan. 3, 2023-Dec. 13, 2022) (217 pages total).

Digital Platform Litigation Documents Part 8, includes IPR2023-00347 (documents filed Dec. 13, 2022), EDTX-2-22-cv-00243 (documents filed Sep. 20, 2022-Jun. 29, 2022), and DDE-1-21-cv-01796 (documents filed Feb. 3, 2023-Jan. 10, 2023 (480 pages total).

Digital Platform Litigation Documents Part 9, includes DDE-1-21-cv-01796 (documents filed Jan. 10, 2023-Nov. 1, 2022 (203 pages total).

Ding et al., "MB2C Model-Based Deep Reinforcement Learning for Multi-zone Building Control," BuildSys '20, Virtual Event, Japan, Nov. 18-20, 2020 (10 pages).

El Kaed, C. et al., "Building management insights driven by a multi-system semantic representation approach," 2016 IEEE 3rd World Forum on Internet of Things (WF-IoT), Dec. 12-14, 2016, (pp. 520-525).

Ellis, C. et al., "Creating a room connectivity graph of a building from per-room sensor units," BuildSys '12, Toronto, ON, Canada, Nov. 6, 2012 (7 pages).

Extended European Search Report on EP Application No. 18196948.6 dated Apr. 10, 2019 (9 pages).

Fierro et al., "Beyond a House of Sticks: Formalizing Metadata Tags with Brick," BuildSys '19, New York, NY, USA, Nov. 13-14, 2019 (10 pages).

Fierro et al., "Dataset: An Open Dataset and Collection Tool for BMS Point Labels," DATA'19, New York, NY, USA, Nov. 10, 2019 (3 pages).

Fierro et al., "Design and Analysis of a Query Processor for Brick," ACM Transactions on Sensor Networks, Jan. 2018, vol. 1, No. 1, art. 1 (25 pages).

Fierro et al., "Design and Analysis of a Query Processor for Brick," BuildSys '17, Delft, Netherlands, Nov. 8-9, 2017 (10 pages).

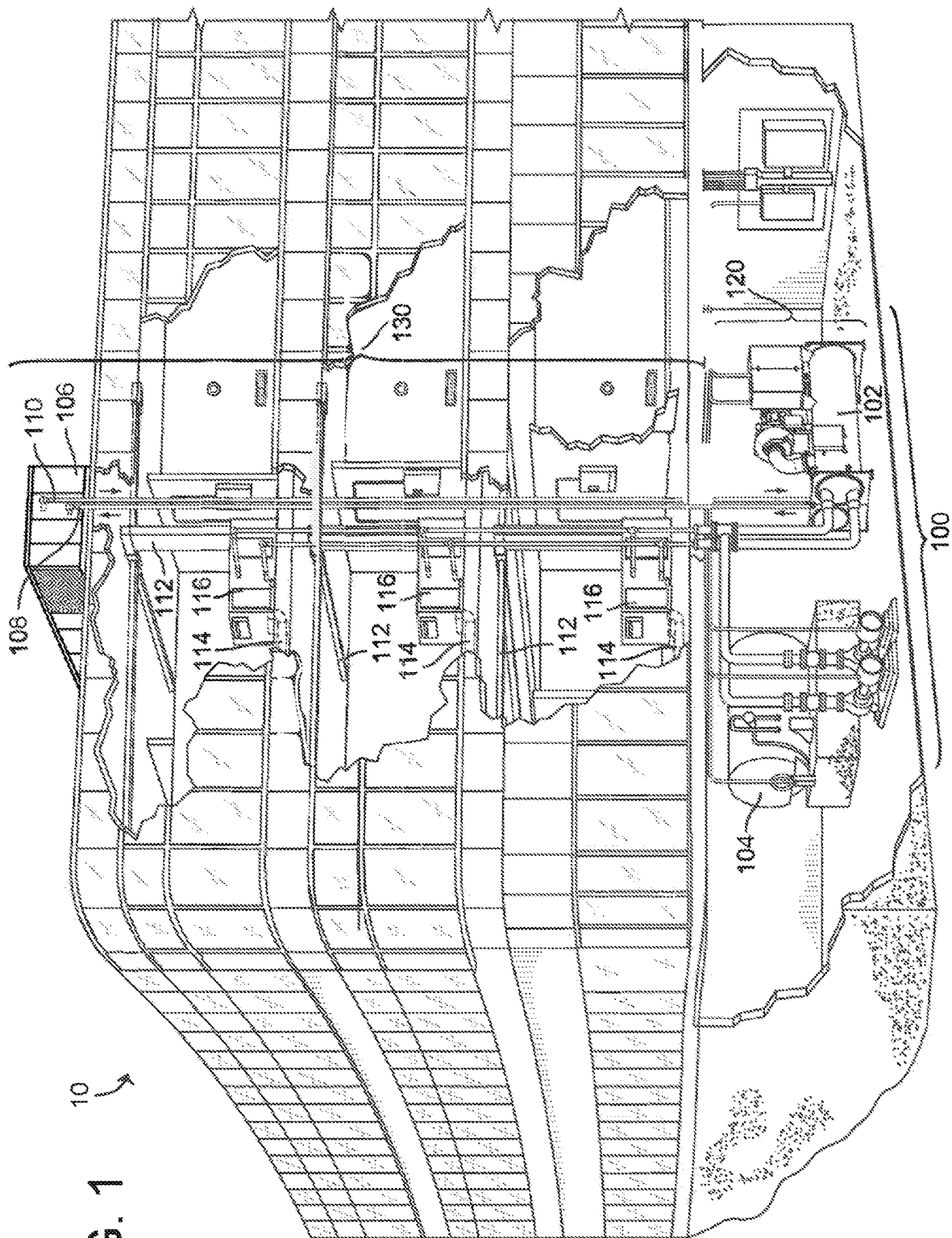
(56)

References Cited

OTHER PUBLICATIONS

- Fierro et al., "Mortar: An Open Testbed for Portable Building Analytics," BuildSys '18, Shenzhen, China, Nov. 7-8, 2018 (10 pages).
- Fierro et al., "Why Brick is a Game Changer for Smart Buildings," URL: https://brickschema.org/papers/Brick_Memoori_Webinar_Presentation.pdf, Memoori Webinar, 2019 (67 pages).
- Fierro, "Writing Portable Building Analytics with the Brick Metadata Schema," UC Berkeley, ACM E-Energy, 2019 (39 pages).
- Fierro, G., "Design of an Effective Ontology and Query Processor Enabling Portable Building Applications," Electrical Engineering and Computer Sciences, University of California at Berkeley, Technical Report No. UCB/EECS-2019-106, June 27, 2019 (118 pages).
- File History for U.S. Appl. No. 12/776,159, filed May 7, 2010 (722 pages).
- Final Conference Program, ACM BuildSys 2016, Stanford, CA, USA, Nov. 15-17, 2016 (7 pages).
- Gao et al., "A large-scale evaluation of automated metadata inference approaches on sensors from air handling units," Advanced Engineering Informatics, 2018, 37 (pp. 14-30).
- Harvey, T., "Quantum Part 3: The Tools of Autonomy, How PassiveLogic's Quantum Creator and Autonomy Studio software works," URL: <https://www.automatedbuildings.com/news/jan22/articles/passive/211224010000passive.html>, Jan. 2022 (7 pages).
- Harvey, T., "Quantum: The Digital Twin Standard for Buildings," URL: <https://www.automatedbuildings.com/news/feb21/articles/passive/210127124501passive.html>, Feb. 2021 (6 pages).
- Hu, S. et al., "Building performance optimisation: A hybrid architecture for the integration of contextual information and time-series data," Automation in Construction, 2016, 70 (pp. 51-61).
- International Search Report and Written Opinion for PCT Appl. Ser. No. PCT/US2017/013831 dated Mar. 31, 2017 (14 pages).
- International Search Report and Written Opinion for PCT Appl. Ser. No. PCT/US2017/035524 dated Jul. 24, 2017 (14 pages).
- International Search Report and Written Opinion on PCT/US2017/052060, mailed Oct. 5, 2017, 11 pages.
- International Search Report and Written Opinion on PCT/US2017/052633, mailed Oct. 23, 2017, 9 pages.
- International Search Report and Written Opinion on PCT/US2017/052829, mailed Nov. 27, 2017, 24 pages.
- International Search Report and Written Opinion on PCT/US2018/024068, mailed Jun. 15, 2018, 22 pages.
- International Search Report and Written Opinion on PCT/US2018/052971, dated Mar. 1, 2019, 19 pages.
- International Search Report and Written Opinion on PCT/US2018/052974, mailed Dec. 19, 2018, 13 pages.
- International Search Report and Written Opinion on PCT/US2018/052975, mailed Jan. 2, 2019, 13 pages.
- International Search Report and Written Opinion on PCT/US2018/052994, mailed Jan. 7, 2019, 15 pages.
- International Search Report and Written Opinion on PCT/US2019/015481, dated May 17, 2019, 78 pages.
- International Search Report and Written Opinion on PCT/US2020/058381, dated Jan. 27, 2021, 30 pages.
- Invitation to Pay Additional Fees and, Where Applicable, Protest Fee on PCT Appl. No. PCT/US2022/050079, dated Feb. 6, 2023 (17 pages).
- Japanese Office Action on JP Appl. No. 2018-534963 dated May 11, 2021 (16 pages with English language translation).
- Koh et al., "Plaster: An Integration, Benchmark, and Development Framework for Metadata Normalization Methods," BuildSys '18, Shenzhen, China, Nov. 7-8, 2018 (10 pages).
- Koh et al., "Scrabble: Transferrable Semi-Automated Semantic Metadata Normalization using Intermediate Representation," BuildSys '18, Shenzhen, China, Nov. 7-8, 2018 (10 pages).
- Koh et al., "Who can Access What, and When?" BuildSys '19, New York, NY, USA, Nov. 13-14, 2019 (4 pages).
- Li et al., "Event Stream Processing with Out-of-Order Data Arrival," International Conferences on Distributed Computing Systems, 2007, (8 pages).
- Nissin Electric Co., Ltd., "Smart power supply system (SPSS)," Outline of the scale verification plan, Nissin Electric Technical Report, Japan, Apr. 23, 2014, vol. 59, No. 1 (23 pages).
- PassiveLogic, "Explorer: Digital Twin Standard for Autonomous Systems. Made interactive." URL: <https://passivelogic.com/software/quantum-explorer/>, retrieved from internet Jan. 4, 2023 (13 pages).
- PassiveLogic, "Quantum: The Digital Twin Standard for Autonomous Systems, A physics-based ontology for next-generation control and AI." URL: <https://passivelogic.com/software/quantum-standard/>, retrieved from internet Jan. 4, 2023 (20 pages).
- Quantum Alliance, "Quantum Explorer Walkthrough," 2022, (7 pages) (screenshots from video).
- Results of the Partial International Search for PCT/US2018/052971, dated Jan. 3, 2019, 3 pages.
- Sinha, Sudhi and Al Huraimel, Khaled, "Reimagining Businesses with AI" John Wiley & Sons, Inc., Hoboken, NJ, USA, First ed. published 2020 (156 pages).
- Sinha, Sudhi R. and PARK, Youngchoon, "Building an Effective IoT Ecosystem for Your Business," Johnson Controls International, Springer International Publishing, 2017 (286 pages).
- Sinha, Sudhi, "Making Big Data Work For Your Business: A guide to effective Big Data analytics," Impact Publishing LTD., Birmingham, UK, Oct. 2014 (170 pages).
- The Virtual Nuclear Tourist, "Calvert Cliffs Nuclear Power Plant," URL: <http://www.nucleartourist.com/us/calvert.htm>, Jan. 11, 2006 (2 pages).
- University of California at Berkeley, EECS Department, "Enabling Scalable Smart-Building Analytics," URL: <https://www2.eecs.berkeley.edu/Pubs/TechRpts/2016/EECS-2016-201.html>, retrieved from internet Feb. 15, 2022 (7 pages).
- Van Hoof, Bert, "Announcing Azure Digital Twins: Create digital replicas of spaces and infrastructure using cloud, AI and IoT," URL: <https://azure.microsoft.com/en-us/blog/announcing-azure-digital-twins-create-digital-replicas-of-spaces-and-infrastructure-using-cloud-ai-and-iot/>, Sep. 24, 2018 (11 pages).
- W3C, "SPARQL: Query Language for RDF," located on The Wayback Machine, URL: <https://web.archive.org/web/20161230061728/http://www.w3.org/TR/rdf-sparql-query/>, retrieved from internet Nov. 15, 2022 (89 pages).
- Wang et al., "Linking energy-cyber-physical systems with occupancy prediction and interpretation through WiFi probe-based ensemble classification," Applied Energy, 2019, 236 (pp. 55-69).
- Wei et al., "Development and Implementation of Software Gateways of Fire Fighting Subsystem Running on EBI," Control, Automation and Systems Engineering, IITA International Conference on, IEEE, Jul. 2009 (pp. 9-12).
- White et al., "Reduce building maintenance costs with AWS IoT TwinMaker Knowledge Graph," The Internet of Things on AWS—Official Blog, URL: <https://aws.amazon.com/blogs/iot/reduce-building-maintenance-costs-with-aws-iot-twinmaker-knowledge-graph/>, Nov. 18, 2022 (10 pages).
- Yao et al., "State of the art review on model predictive control (MPC) in Heating Ventilation and Air-conditioning (HVAC) field," Building and Environment, 2021, 200 (18 pages).
- Zhou, Q. et al., "Knowledge-infused and Consistent Complex Event Processing over Real-time and Persistent Streams," Further Generation Computer Systems, 2017, 76 (pp. 391-406).
- U.S. Appl. No. 17/529,118, filed Nov. 17, 2021.
- U.S. Appl. No. 17/566,029, PassiveLogic, Inc.
- U.S. Appl. No. 17/567,275, PassiveLogic, Inc.
- U.S. Appl. No. 17/722,115, PassiveLogic, Inc.
- Sjarov et al., "The Digital Twin Concept in Industry—A Review and Systemization," 2020 25th IEEE International Conference on Emerging Technologies and Factory Automation (EFTA), Sep. 8-11, 2020 (8 pages).

* cited by examiner



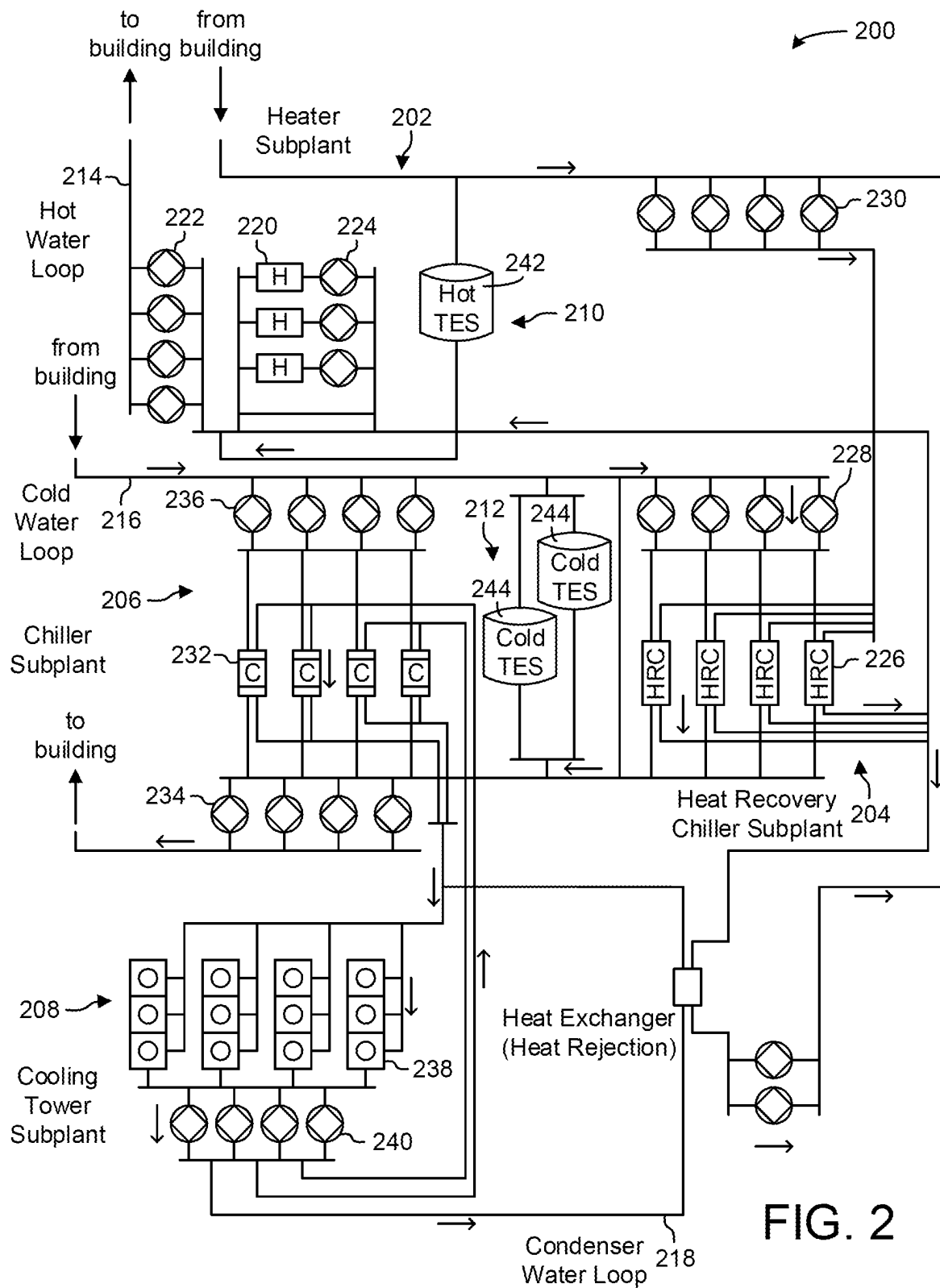


FIG. 2

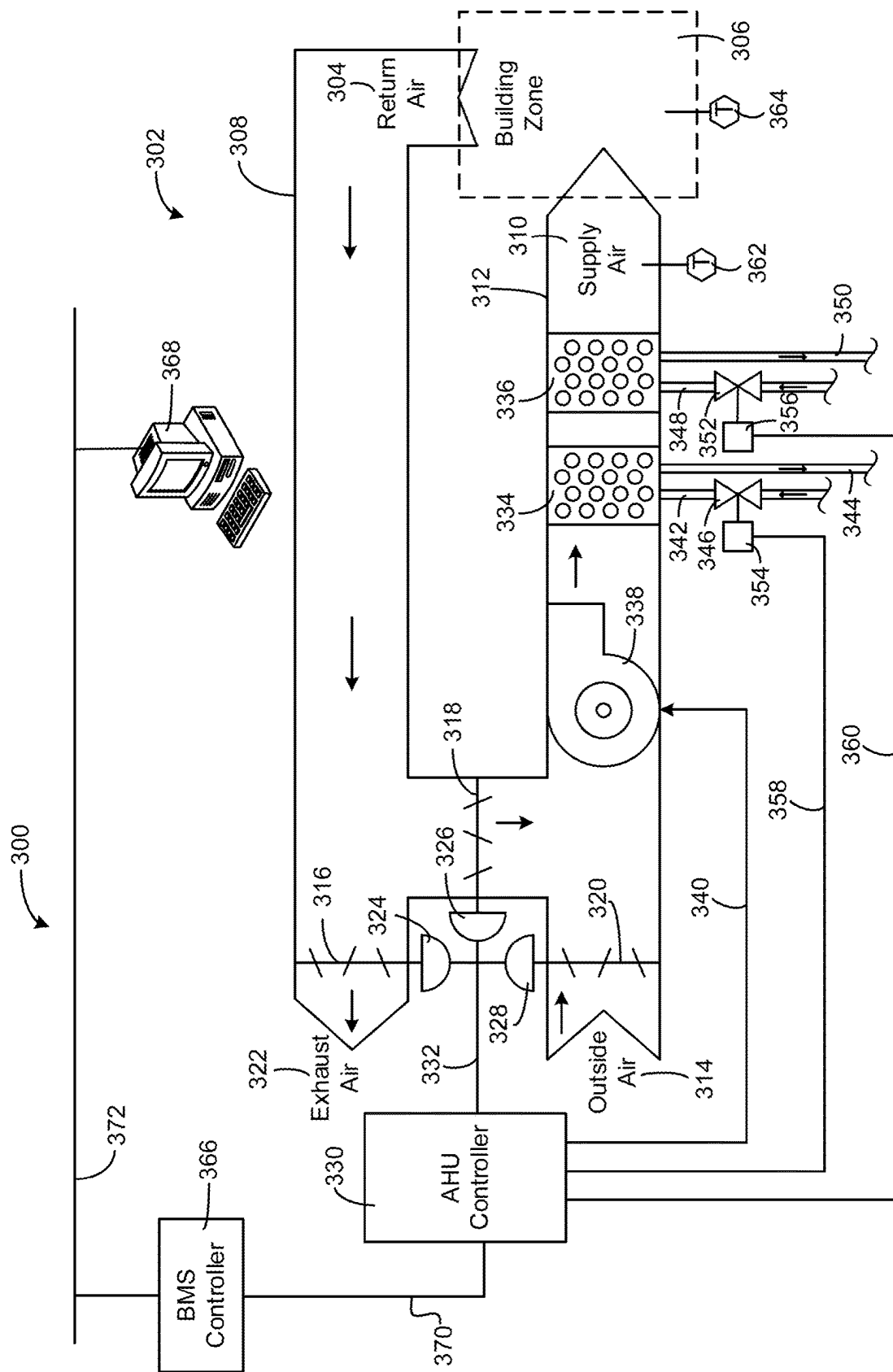


FIG. 3

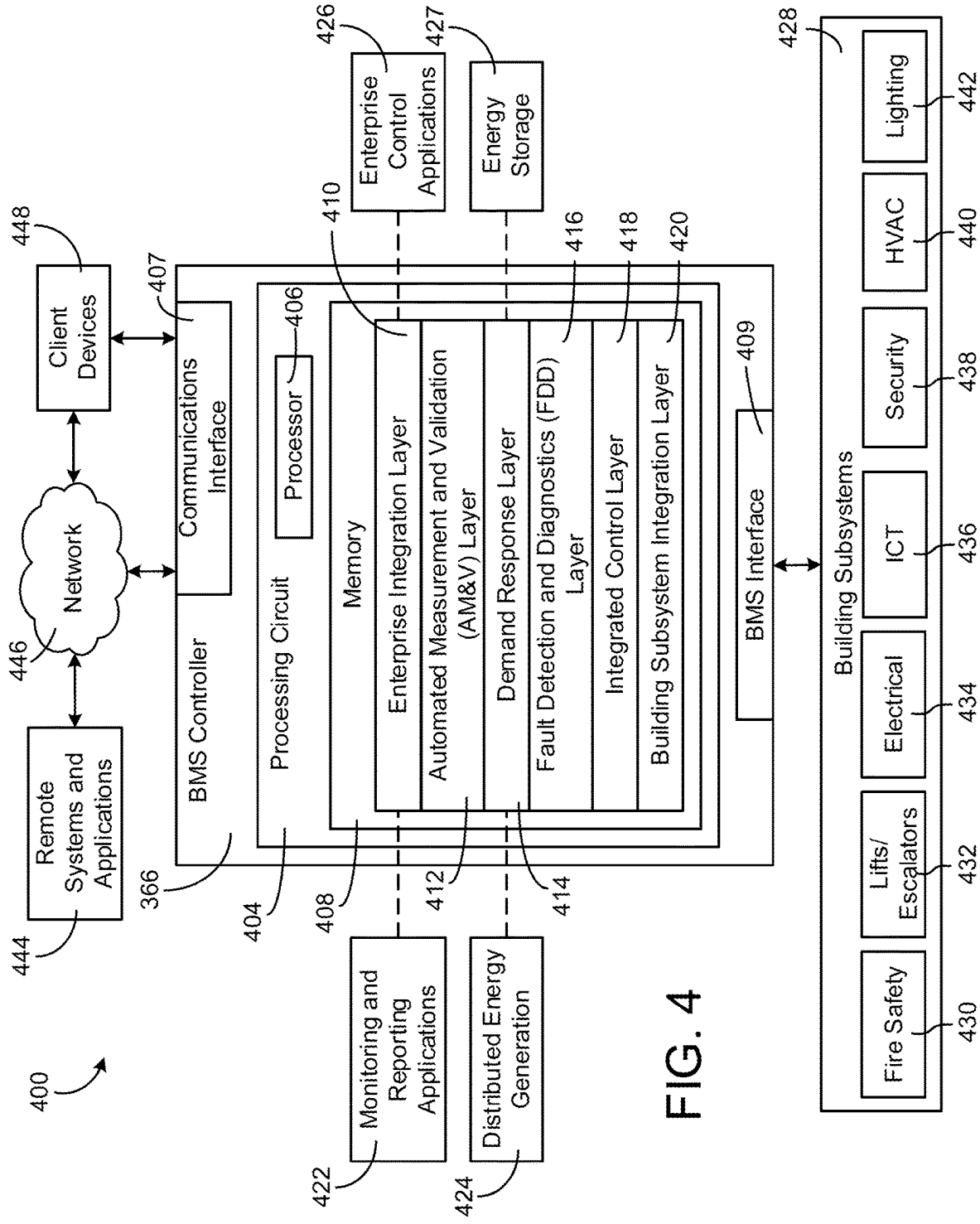


FIG. 4

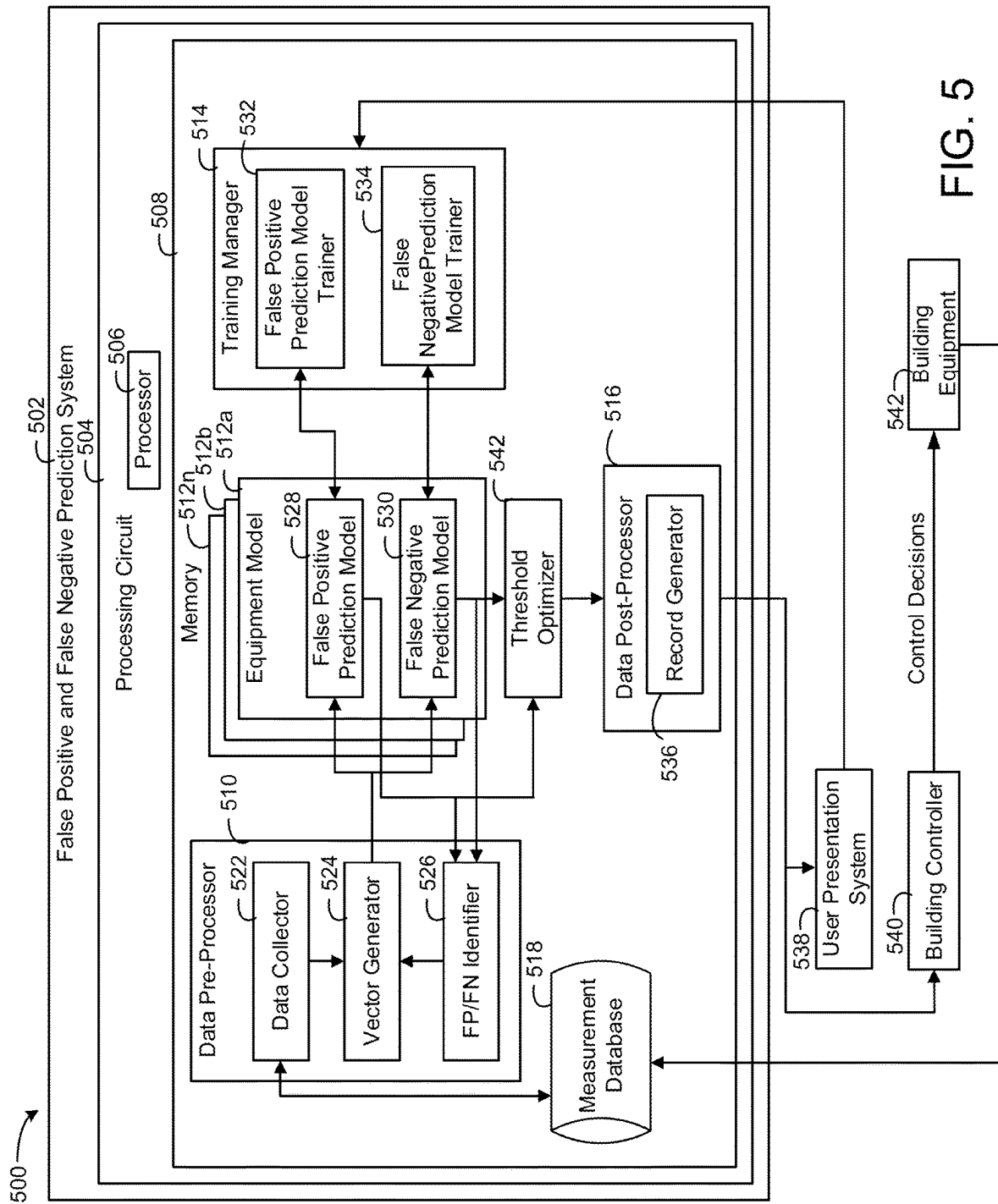


FIG. 5

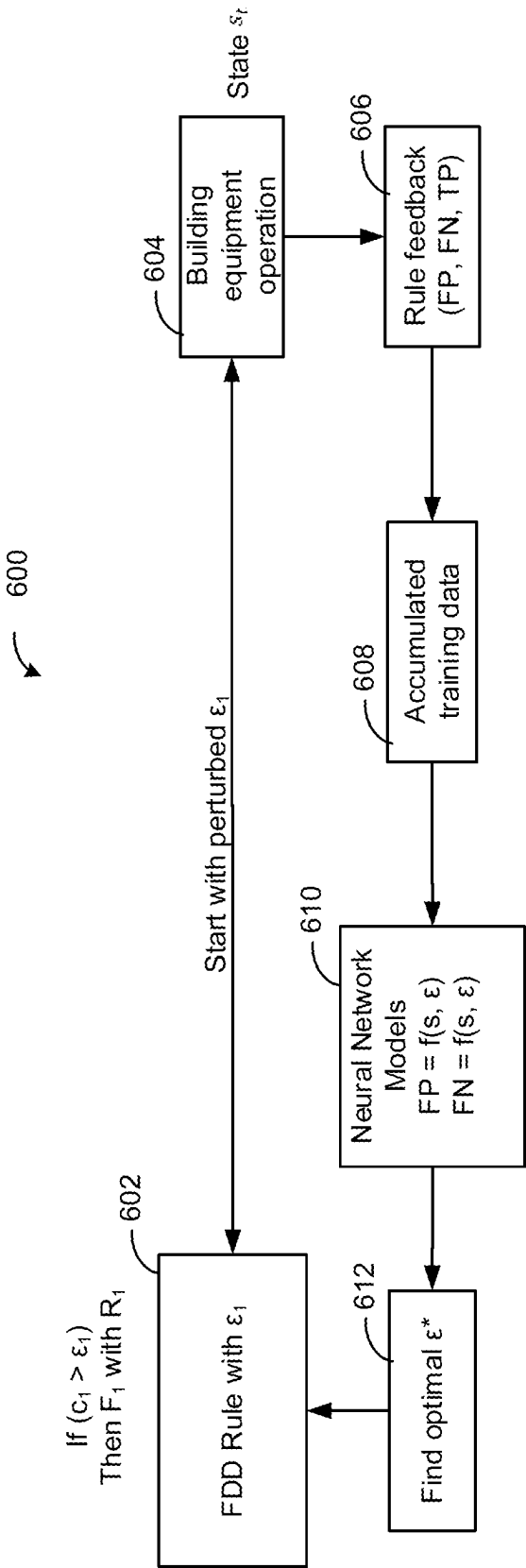


FIG. 6

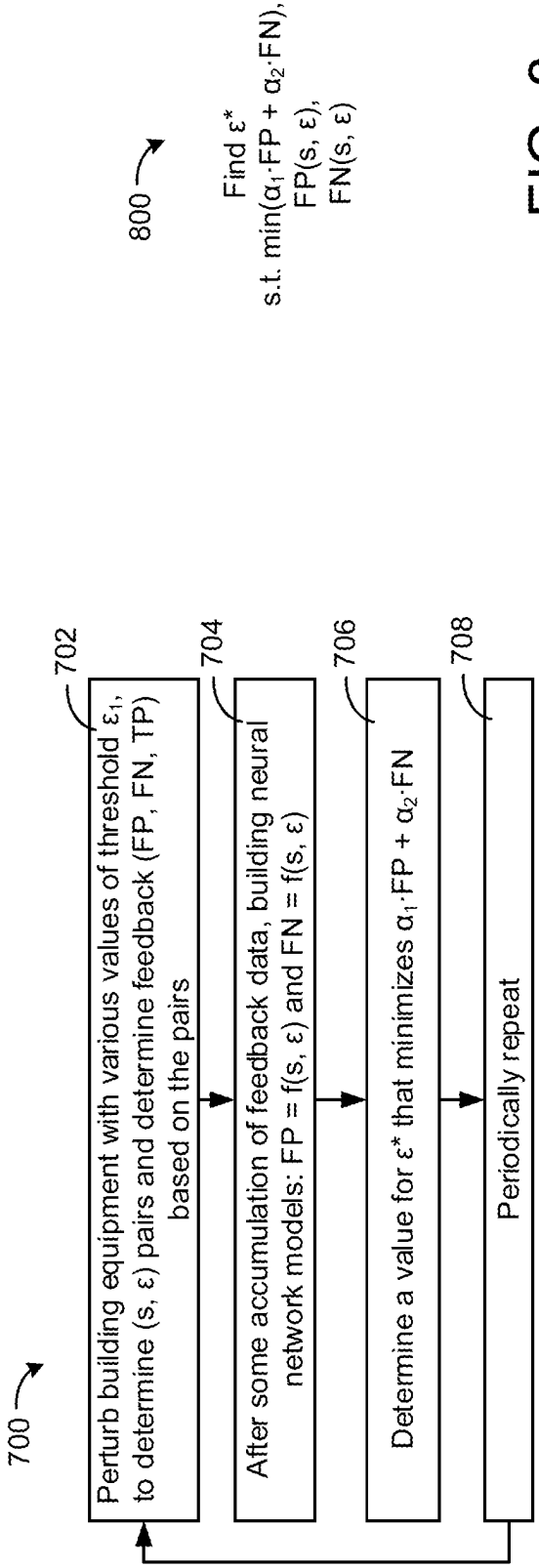


FIG. 7

800 →

$$\begin{aligned} &\text{Find } \epsilon^* \\ &\text{s.t. } \min(\alpha_1 \cdot FP + \alpha_2 \cdot FN), \\ &\quad FP(s, \epsilon), \\ &\quad FN(s, \epsilon) \end{aligned}$$

FIG. 8

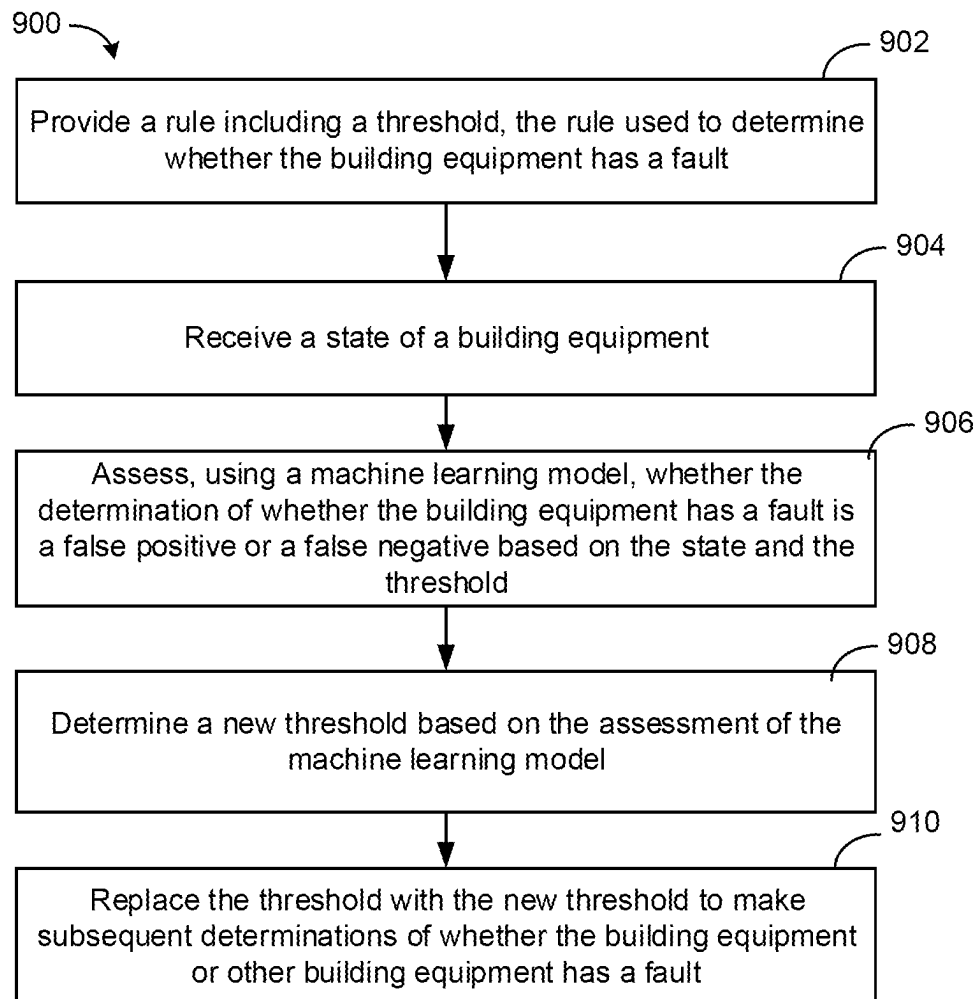


FIG. 9

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BUILDING MANAGEMENT SYSTEMS AND METHODS FOR TUNING FAULT DETECTION THRESHOLDS

CROSS-REFERENCE TO RELATED PATENT APPLICATIONS

The present application is a continuation-in-part of U.S. patent application Ser. No. 17/529,118, filed Nov. 17, 2021, which is incorporated herein by reference in its entirety.

BACKGROUND

This application relates generally to a building system of a building. This application relates more particularly to systems for tuning fault detection of the building system.

A building management system (BMS) is, in general, a system of devices configured to control, monitor, and manage equipment in or around a building or building area. A BMS can include, for example, a HVAC system, a security system, a lighting system, a fire alerting system, any other system that is capable of managing building functions or devices, or any combination thereof. Specifically, the present disclosure relates to a BMS with an equipment monitoring system to accurately determine whether the BMS is experiencing a fault.

SUMMARY

One inventive aspect is a non-transitory computer-readable storage medium having instructions stored thereon that, when executed by one or more processors, cause the one or more processors to provide a rule including a threshold, the rule used to determine whether building equipment has a fault. The instructions further cause the one or more processors to receive a state of the building equipment, assess, using a machine learning model, whether the determination of whether the building equipment has a fault is a false positive or a false negative based on the state and the threshold, determine a new threshold based on the assessment of the machine learning model, and replace the threshold with the new threshold to make subsequent determinations of whether the building equipment or other building equipment has a fault.

In some embodiments, the new threshold reduces a number of false positives a number of false negatives.

In some embodiments, the instructions further cause the one or more processors to accumulate training data for the machine learning model using the state and the threshold as inputs and labels of false positive or false negative as outputs.

In some embodiments, the rule includes a condition portion including a comparison of the state to the threshold and an action portion including a determination that the building equipment is faulty when the condition is satisfied or not faulty when the condition is not satisfied.

In some embodiments, the machine learning model includes a first machine learning model for predicting a false negative and a second machine learning model for predicting a false positive.

In some embodiments, the instructions further cause the one or more processors to perturb the building equipment with multiple values of the threshold to provide additional data for the machine learning model.

In some embodiments, the first machine learning model is configured to accurately predict false positives at a first rate,

2

and the instructions further cause the one or more processors to retrain the first machine learning model when the first rate exceeds a first threshold rate.

In some embodiments, the second machine learning model is configured accurately predict false negatives at a second rate, and the instructions further cause the one or more processors to retrain the second machine learning model when the second rate exceeds a second threshold rate.

In some embodiments, the one or more processors are configured to use a constrained nonlinear optimization to determine the new threshold.

Another aspect is a method including providing, by a processing circuit, a rule including a threshold, the rule used to determine whether building equipment has a fault. The method includes receiving, by the processing circuit, a state of the building equipment, assessing, by the processing circuit using a machine learning model, whether the determination of whether the building equipment has a fault is a false positive or a false negative based on the state and the threshold, determining, by the processing circuit, a new threshold based on the assessment of the machine learning model, and replacing, by the processing circuit, the threshold with the new threshold to make subsequent determinations of whether the building equipment or other building equipment has a fault.

In some embodiments, the new threshold reduces a number of false positives a number of false negatives.

In some embodiments, accumulating, by the processing circuit, training data for the machine learning model using the state and the threshold as inputs and labels of false positive or false negative as outputs.

In some embodiments, the rule includes a condition portion including a comparison of the state to the threshold and an action portion including a determination that the building equipment is faulty when the condition is satisfied or not faulty when the condition is not satisfied.

In some embodiments, the machine learning model includes a first machine learning model for predicting a false negative and a second machine learning model for predicting a false positive.

In some embodiments, perturbing, by the processing circuit, the building equipment with multiple values of the threshold to provide additional data for the machine learning model.

In some embodiments, the first machine learning model is configured to accurately predict false positives at a first rate, and the method further includes retraining, by the processing circuit, the first machine learning model when the first rate exceeds a first threshold rate.

In some embodiments, the second machine learning model is configured accurately predict false negatives at a second rate, and the method further includes retraining, by the processing circuit, the second machine learning model when the second rate exceeds a second threshold rate.

In some embodiments, the method further includes using, by the processing circuit, a constrained nonlinear optimization to determine the new threshold.

Another aspect is a building system including one or more storage devices storing instructions thereon and one or more processors which execute the instructions causing the one or more processors to provide a rule used to determine whether a building equipment has a fault, perturb the building equipment having a plurality of states with a plurality of corresponding thresholds for the rule for determining whether a fault exists, determine whether a fault exists based on the perturbed building equipment, and receive feedback of whether the determination of whether a fault exists is a

false positive or a false negative, or a true positive. The one or more processors further execute the instructions causing the one or more processors to provide training data to a machine learning model, the training data including the plurality of states and the plurality of corresponding thresholds as inputs and the feedback of false positive or false negative as outputs. The one or more processors further execute the instructions causing the one or more processors to receive a current state of the building equipment and a current threshold of the rule, assess, using the trained machine learning model, whether the determination of whether the building equipment has a fault is a false positive or a false negative based on the current state and the current threshold, determine a new threshold based on the assessment of the trained machine learning model, and replace the current threshold with the new threshold to make subsequent determinations of whether the building equipment or other building equipment has a fault.

In some embodiments, the instructions further cause the one or more processors to use constrained nonlinear optimization to find the new threshold that reduces a number of false positives a number of false negatives.

BRIEF DESCRIPTION OF THE DRAWINGS

Various objects, aspects, features, and advantages of the disclosure will become more apparent and better understood by referring to the detailed description taken in conjunction with the accompanying drawings, in which like reference characters identify corresponding elements throughout. In the drawings, like reference numbers generally indicate identical, functionally similar, and/or structurally similar elements.

FIG. 1 is a drawing of a building equipped with a HVAC system, according to an exemplary embodiment.

FIG. 2 is a block diagram of a waterside system that may be used in conjunction with the building of FIG. 1, according to an exemplary embodiment.

FIG. 3 is a block diagram of an airside system that may be used in conjunction with the building of FIG. 1, according to an exemplary embodiment.

FIG. 4 is a block diagram of a building automation system (BAS) that may be used to monitor and/or control the building of FIG. 1, according to an exemplary embodiment.

FIG. 5 is a block diagram of a system including a fault detection system, according to some embodiments.

FIG. 6 is a block diagram of a system to find an optimal threshold of a fault detection and diagnosis rule, according to an exemplary embodiment.

FIG. 7 is a flow diagram of a process for identifying values for the parameters of the threshold of FIG. 6, according to an exemplary embodiment.

FIG. 8 is an optimization algorithm that can be performed to identify the optimal threshold for the fault detection and diagnosis rule of FIGS. 6 and 7, according to an exemplary embodiment.

FIG. 9 is a flow diagram of a process for calculating the optimal threshold, according to an exemplary embodiment.

DETAILED DESCRIPTION

Overview

Referring generally to the FIGURES, systems and methods for autonomous fault detection and diagnosis rule threshold tuning are shown, according to various exemplary embodiments. One method of detecting faults in buildings

and their equipment is using rules to detect and diagnose faults. For example, when a building controller detects that a rule is satisfied, the building controller may determine that a subsystem pertaining to or associated with the rule has or may have a fault that should be adjusted and/or fixed.

A common technique of detecting faults in building equipment is using rules within building management systems (BMS). There may be one or more rules pertaining any particular building equipment (e.g., air handling unit, lighting equipment, etc.). Ideally the BMS will always correctly determine whether the building equipment is experiencing a fault or not. However, it is possible that the BMS determines that a fault exists even when the building equipment is operating normally (false positive), or that the BMS determines that a fault does not exist even when the building equipment does actually have a fault (false negative). When a fault is detected, the building manager may be notified by an alarm so that the building equipment may be inspected and/or a work order may be generated for a technician to fix the building equipment. This failure in accurate determination can lead to equipment failures, occupant discomfort and/or safety concerns, excessive costs, and waste.

Rules often include a condition portion and an action portion. When the condition portion is satisfied, the action portion may be executed. For example, there may be a rule for an air handling unit (AHU) that states, in the condition portion, that the supply air temperature is below a first threshold temperature and the fan speed of the AHU is greater than a second threshold rotations per minute (rpm). The rule may have an action portion that states that the AHU is at fault. In this rule, the condition portion is satisfied if both the supply air temperature is below the first threshold and the fan speed is greater than the second threshold, and the system can receive a determination that the AHU is faulty.

Rules are not always set for every condition and for every equipment. Accordingly, the rules may need to be adjusted depending on operating conditions, equipment, occupants, etc. Further, the rules may need to be adjusted because the equipment may degrade over time. If a threshold is too small, normal variation in operating conditions may result in false alarms, and if the threshold is too great, only a few extremely severe faults may be detected. Furthermore, manually adjusting or tuning thresholds is very difficult because of inaccuracies or overcompensation, which can lead to similar or different problems. Accordingly, there is a need to automatically tune the thresholds in rules.

In the present disclosure, systems and methods of automatically tuning the thresholds in rules are described. First, some data needs to be collected concerning the rules and whether the rules resulted in false positives or false negatives. The data can include the state of the building equipment, the threshold of the rule, and a determination of whether the rule resulted in a false positive or a false negative. This data can be provided as training data for a machine learning model or multiple models that can be trained to predict when a false positive or a false negative is going to occur based on the state and threshold. Once the model is trained, an optimization technique can be used to determine the optimal threshold level that minimizes the numbers of false positives and false negatives. The optimal threshold can then replace the existing threshold so that false positives and false negatives are minimized.

Building Management System Overview

Referring now to FIGS. 1-4, an exemplary building automation system (BAS) and HVAC system in which the systems and methods of the present invention can be imple-

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mented are shown, according to an exemplary embodiment. Referring particularly to FIG. 1, a perspective view of a building 10 is shown. Building 10 is served by a BAS. A BAS is, in general, a system of devices configured to control, monitor, and manage equipment in or around a building or building area. A BAS can include, for example, a HVAC system, a security system, a lighting system, a fire alarming system, any other system that is capable of managing building functions or devices, or any combination thereof.

The BAS that serves building 10 includes an HVAC system 100. HVAC system 100 can include a plurality of HVAC devices (e.g., heaters, chillers, air handling units, pumps, fans, thermal energy storage, etc.) configured to provide heating, cooling, ventilation, or other services for building 10. For example, HVAC system 100 is shown to include a waterside system 120 and an airside system 130. Waterside system 120 can provide a heated or chilled fluid to an air handling unit of airside system 130. Airside system 130 can use the heated or chilled fluid to heat or cool an airflow provided to building 10. An exemplary waterside system and airside system which can be used in HVAC system 100 are described in greater detail with reference to FIGS. 2-3.

HVAC system 100 is shown to include a chiller 102, a boiler 104, and a rooftop air handling unit (AHU) 106. Waterside system 120 can use boiler 104 and chiller 102 to heat or cool a working fluid (e.g., water, glycol, etc.) and can circulate the working fluid to AHU 106. In various embodiments, the HVAC devices of waterside system 120 can be located in or around building 10 (as shown in FIG. 1) or at an offsite location such as a central plant (e.g., a chiller plant, a steam plant, a heat plant, etc.). The working fluid can be heated in boiler 104 or cooled in chiller 102, depending on whether heating or cooling is required in building 10. Boiler 104 can add heat to the circulated fluid, for example, by burning a combustible material (e.g., natural gas) or using an electric heating element. Chiller 102 can place the circulated fluid in a heat exchange relationship with another fluid (e.g., a refrigerant) in a heat exchanger (e.g., an evaporator) to absorb heat from the circulated fluid. The working fluid from chiller 102 and/or boiler 104 can be transported to AHU 106 via piping 108.

AHU 106 can place the working fluid in a heat exchange relationship with an airflow passing through AHU 106 (e.g., via one or more stages of cooling coils and/or heating coils). The airflow can be, for example, outside air, return air from within building 10, or a combination of both. AHU 106 can transfer heat between the airflow and the working fluid to provide heating or cooling for the airflow. For example, AHU 106 can include one or more fans or blowers configured to pass the airflow over or through a heat exchanger containing the working fluid. The working fluid can then return to chiller 102 or boiler 104 via piping 110.

Airside system 130 can deliver the airflow supplied by AHU 106 (i.e., the supply airflow) to building 10 via air supply ducts 112 and can provide return air from building 10 to AHU 106 via air return ducts 114. In some embodiments, airside system 130 includes multiple variable air volume (VAV) units 116. For example, airside system 130 is shown to include a separate VAV unit 116 on each floor or zone of building 10. VAV units 116 can include dampers or other flow control elements that can be operated to control an amount of the supply airflow provided to individual zones of building 10. In other embodiments, airside system 130 delivers the supply airflow into one or more zones of building 10 (e.g., via supply ducts 112) without using

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intermediate VAV units 116 or other flow control elements. AHU 106 can include various sensors (e.g., temperature sensors, pressure sensors, etc.) configured to measure attributes of the supply airflow. AHU 106 can receive input from sensors located within AHU 106 and/or within the building zone and can adjust the flow rate, temperature, or other attributes of the supply airflow through AHU 106 to achieve setpoint conditions for the building zone.

Referring now to FIG. 2, a block diagram of a waterside system 200 is shown, according to an exemplary embodiment. In various embodiments, waterside system 200 can supplement or replace waterside system 120 in HVAC system 100 or can be implemented separate from HVAC system 100. When implemented in HVAC system 100, waterside system 200 can include a subset of the HVAC devices in HVAC system 100 (e.g., boiler 104, chiller 102, pumps, valves, etc.) and can operate to supply a heated or chilled fluid to AHU 106. The HVAC devices of waterside system 200 can be located within building 10 (e.g., as components of waterside system 120) or at an offsite location such as a central plant.

In FIG. 2, waterside system 200 is shown as a central plant having a plurality of subplants 202-212. Subplants 202-212 are shown to include a heater subplant 202, a heat recovery chiller subplant 204, a chiller subplant 206, a cooling tower subplant 208, a hot thermal energy storage (TES) subplant 210, and a cold thermal energy storage (TES) subplant 212. Subplants 202-212 consume resources (e.g., water, natural gas, electricity, etc.) from utilities to serve the thermal energy loads (e.g., hot water, cold water, heating, cooling, etc.) of a building or campus. For example, heater subplant 202 can be configured to heat water in a hot water loop 214 that circulates the hot water between heater subplant 202 and building 10. Chiller subplant 206 can be configured to chill water in a cold water loop 216 that circulates the cold water between chiller subplant 206 and building 10. Heat recovery chiller subplant 204 can be configured to transfer heat from cold water loop 216 to hot water loop 214 to provide additional heating for the hot water and additional cooling for the cold water. Condenser water loop 218 can absorb heat from the cold water in chiller subplant 206 and reject the absorbed heat in cooling tower subplant 208 or transfer the absorbed heat to hot water loop 214. Hot TES subplant 210 and cold TES subplant 212 can store hot and cold thermal energy, respectively, for subsequent use.

Hot water loop 214 and cold water loop 216 can deliver the heated and/or chilled water to air handlers located on the rooftop of building 10 (e.g., AHU 106) or to individual floors or zones of building 10 (e.g., VAV units 116). The air handlers push air past heat exchangers (e.g., heating coils or cooling coils) through which the water flows to provide heating or cooling for the air. The heated or cooled air can be delivered to individual zones of building 10 to serve the thermal energy loads of building 10. The water then returns to subplants 202-212 to receive further heating or cooling.

Although subplants 202-212 are shown and described as heating and cooling water for circulation to a building, it is understood that any other type of working fluid (e.g., glycol, CO₂, etc.) can be used in place of or in addition to water to serve the thermal energy loads. In other embodiments, subplants 202-212 can provide heating and/or cooling directly to the building or campus without requiring an intermediate heat transfer fluid. These and other variations to waterside system 200 are within the teachings of the present invention.

Each of subplants **202-212** can include a variety of equipment configured to facilitate the functions of the subplant. For example, heater subplant **202** is shown to include a plurality of heating elements **220** (e.g., boilers, electric heaters, etc.) configured to add heat to the hot water in hot water loop **214**. Heater subplant **202** is also shown to include several pumps **222** and **224** configured to circulate the hot water in hot water loop **214** and to control the flow rate of the hot water through individual heating elements **220**. Chiller subplant **206** is shown to include a plurality of chillers **232** configured to remove heat from the cold water in cold water loop **216**. Chiller subplant **206** is also shown to include several pumps **234** and **236** configured to circulate the cold water in cold water loop **216** and to control the flow rate of the cold water through individual chillers **232**.

Heat recovery chiller subplant **204** is shown to include a plurality of heat recovery heat exchangers **226** (e.g., refrigeration circuits) configured to transfer heat from cold water loop **216** to hot water loop **214**. Heat recovery chiller subplant **204** is also shown to include several pumps **228** and **230** configured to circulate the hot water and/or cold water through heat recovery heat exchangers **226** and to control the flow rate of the water through individual heat recovery heat exchangers **226**. Cooling tower subplant **208** is shown to include a plurality of cooling towers **238** configured to remove heat from the condenser water in condenser water loop **218**. Cooling tower subplant **208** is also shown to include several pumps **240** configured to circulate the condenser water in condenser water loop **218** and to control the flow rate of the condenser water through individual cooling towers **238**.

Hot TES subplant **210** is shown to include a hot TES tank **242** configured to store the hot water for later use. Hot TES subplant **210** can also include one or more pumps or valves configured to control the flow rate of the hot water into or out of hot TES tank **242**. Cold TES subplant **212** is shown to include cold TES tanks **244** configured to store the cold water for later use. Cold TES subplant **212** can also include one or more pumps or valves configured to control the flow rate of the cold water into or out of cold TES tanks **244**.

In some embodiments, one or more of the pumps in waterside system **200** (e.g., pumps **222**, **224**, **228**, **230**, **234**, **236**, and/or **240**) or pipelines in waterside system **200** include an isolation valve associated therewith. Isolation valves can be integrated with the pumps or positioned upstream or downstream of the pumps to control the fluid flows in waterside system **200**. In various embodiments, waterside system **200** can include more, fewer, or different types of devices and/or subplants based on the particular configuration of waterside system **200** and the types of loads served by waterside system **200**.

Referring now to FIG. 3, a block diagram of an airside system **300** is shown, according to an exemplary embodiment. In various embodiments, airside system **300** can supplement or replace airside system **130** in HVAC system **100** or can be implemented separate from HVAC system **100**. When implemented in HVAC system **100**, airside system **300** can include a subset of the HVAC devices in HVAC system **100** (e.g., AHU **106**, VAV units **116**, ducts **112-114**, fans, dampers, etc.) and can be located in or around building **10**. Airside system **300** can operate to heat or cool an airflow provided to building **10** using a heated or chilled fluid provided by waterside system **200**.

In FIG. 3, airside system **300** is shown to include an economizer-type air handling unit (AHU) **302**. Economizer-type AHUs vary the amount of outside air and return air used by the air handling unit for heating or cooling. For example,

AHU **302** can receive return air **304** from building zone **306** via return air duct **308** and can deliver supply air **310** to building zone **306** via supply air duct **312**. In some embodiments, AHU **302** is a rooftop unit located on the roof of building **10** (e.g., AHU **106** as shown in FIG. 1) or otherwise positioned to receive both return air **304** and outside air **314**. AHU **302** can be configured to operate exhaust air damper **316**, mixing damper **318**, and outside air damper **320** to control an amount of outside air **314** and return air **304** that combine to form supply air **310**. Any return air **304** that does not pass through mixing damper **318** can be exhausted from AHU **302** through exhaust damper **316** as exhaust air **322**.

Each of dampers **316-320** can be operated by an actuator. For example, exhaust air damper **316** can be operated by actuator **324**, mixing damper **318** can be operated by actuator **326**, and outside air damper **320** can be operated by actuator **328**. Actuators **324-328** can communicate with an AHU controller **330** via a communications link **332**. Actuators **324-328** can receive control signals from AHU controller **330** and can provide feedback signals to AHU controller **330**. Feedback signals can include, for example, an indication of a current actuator or damper position, an amount of torque or force exerted by the actuator, diagnostic information (e.g., results of diagnostic tests performed by actuators **324-328**), status information, commissioning information, configuration settings, calibration data, and/or other types of information or data that can be collected, stored, or used by actuators **324-328**. AHU controller **330** can be an economizer controller configured to use one or more control algorithms (e.g., state-based algorithms, extremum seeking control (ESC) algorithms, proportional-integral (PI) control algorithms, proportional-integral-derivative (PID) control algorithms, model predictive control (MPC) algorithms, feedback control algorithms, etc.) to control actuators **324-328**.

Still referring to FIG. 3, AHU **302** is shown to include a cooling coil **334**, a heating coil **336**, and a fan **338** positioned within supply air duct **312**. Fan **338** can be configured to force supply air **310** through cooling coil **334** and/or heating coil **336** and provide supply air **310** to building zone **306**. AHU controller **330** can communicate with fan **338** via communications link **340** to control a flow rate of supply air **310**. In some embodiments, AHU controller **330** controls an amount of heating or cooling applied to supply air **310** by modulating a speed of fan **338**.

Cooling coil **334** can receive a chilled fluid from waterside system **200** (e.g., from cold water loop **216**) via piping **342** and can return the chilled fluid to waterside system **200** via piping **344**. Valve **346** can be positioned along piping **342** or piping **344** to control a flow rate of the chilled fluid through cooling coil **334**. In some embodiments, cooling coil **334** includes multiple stages of cooling coils that can be independently activated and deactivated (e.g., by AHU controller **330**, by BAS controller **366**, etc.) to modulate an amount of cooling applied to supply air **310**.

Heating coil **336** can receive a heated fluid from waterside system **200** (e.g., from hot water loop **214**) via piping **348** and can return the heated fluid to waterside system **200** via piping **350**. Valve **352** can be positioned along piping **348** or piping **350** to control a flow rate of the heated fluid through heating coil **336**. In some embodiments, heating coil **336** includes multiple stages of heating coils that can be independently activated and deactivated (e.g., by AHU controller **330**, by BAS controller **366**, etc.) to modulate an amount of heating applied to supply air **310**.

Each of valves **346** and **352** can be controlled by an actuator. For example, valve **346** can be controlled by

actuator 354 and valve 352 can be controlled by actuator 356. Actuators 354-356 can communicate with AHU controller 330 via communications links 358-360. Actuators 354-356 can receive control signals from AHU controller 330 and can provide feedback signals to controller 330. In some embodiments, AHU controller 330 receives a measurement of the supply air temperature from a temperature sensor 362 positioned in supply air duct 312 (e.g., downstream of cooling coil 334 and/or heating coil 336). AHU controller 330 can also receive a measurement of the temperature of building zone 306 from a temperature sensor 364 located in building zone 306.

In some embodiments, AHU controller 330 operates valves 346 and 352 via actuators 354-356 to modulate an amount of heating or cooling provided to supply air 310 (e.g., to achieve a setpoint temperature for supply air 310 or to maintain the temperature of supply air 310 within a setpoint temperature range). The positions of valves 346 and 352 affect the amount of heating or cooling provided to supply air 310 by cooling coil 334 or heating coil 336 and may correlate with the amount of energy consumed to achieve a desired supply air temperature. AHU controller 330 can control the temperature of supply air 310 and/or building zone 306 by activating or deactivating coils 334-336, adjusting a speed of fan 338, or a combination of both.

Still referring to FIG. 3, airside system 300 is shown to include a building automation system (BAS) controller 366 and a client device 368. BAS controller 366 can include one or more computer systems (e.g., servers, supervisory controllers, subsystem controllers, etc.) that serve as system level controllers, application or data servers, head nodes, or master controllers for airside system 300, waterside system 200, HVAC system 100, and/or other controllable systems that serve building 10. BAS controller 366 can communicate with multiple downstream building systems or subsystems (e.g., HVAC system 100, a security system, a lighting system, waterside system 200, etc.) via a communications link 370 according to like or disparate protocols (e.g., LON, BACnet, etc.). In various embodiments, AHU controller 330 and BAS controller 366 can be separate (as shown in FIG. 3) or integrated. In an integrated implementation, AHU controller 330 can be a software module configured for execution by a processor of BAS controller 366.

In some embodiments, AHU controller 330 receives information from BAS controller 366 (e.g., commands, setpoints, operating boundaries, etc.) and provides information to BAS controller 366 (e.g., temperature measurements, valve or actuator positions, operating statuses, diagnostics, etc.). For example, AHU controller 330 can provide BAS controller 366 with temperature measurements from temperature sensors 362-364, equipment on/off states, equipment operating capacities, and/or any other information that can be used by BAS controller 366 to monitor or control a variable state or condition within building zone 306.

Client device 368 can include one or more human-machine interfaces or client interfaces (e.g., graphical user interfaces, reporting interfaces, text-based computer interfaces, client-facing web services, web servers that provide pages to web clients, etc.) for controlling, viewing, or otherwise interacting with HVAC system 100, its subsystems, and/or devices. Client device 368 can be a computer workstation, a client terminal, a remote or local interface, or any other type of user interface device. Client device 368 can be a stationary terminal or a mobile device. For example, client device 368 can be a desktop computer, a computer server with a user interface, a laptop computer, a tablet, a smartphone, a PDA, or any other type of mobile or non-

mobile device. Client device 368 can communicate with BAS controller 366 and/or AHU controller 330 via communications link 372.

Referring now to FIG. 4, a block diagram of a building automation system (BAS) 400 is shown, according to an exemplary embodiment. BAS 400 can be implemented in building 10 to automatically monitor and control various building functions. BAS 400 is shown to include BAS controller 366 and a plurality of building subsystems 428. Building subsystems 428 are shown to include a building electrical subsystem 434, an information communication technology (ICT) subsystem 436, a security subsystem 438, a HVAC subsystem 440, a lighting subsystem 442, a lift/escalators subsystem 432, and a fire safety subsystem 430. In various embodiments, building subsystems 428 can include fewer, additional, or alternative subsystems. For example, building subsystems 428 can also or alternatively include a refrigeration subsystem, an advertising or signage subsystem, a cooking subsystem, a vending subsystem, a printer or copy service subsystem, or any other type of building subsystem that uses controllable equipment and/or sensors to monitor or control building 10. In some embodiments, building subsystems 428 include waterside system 200 and/or airside system 300, as described with reference to FIGS. 2-3.

Each of building subsystems 428 can include any number of devices, controllers, and connections for completing its individual functions and control activities. HVAC subsystem 440 can include many of the same components as HVAC system 100, as described with reference to FIGS. 1-3. For example, HVAC subsystem 440 can include a chiller, a boiler, any number of air handling units, economizers, field controllers, supervisory controllers, actuators, temperature sensors, and other devices for controlling the temperature, humidity, airflow, or other variable conditions within building 10. Lighting subsystem 442 can include any number of light fixtures, ballasts, lighting sensors, dimmers, or other devices configured to controllably adjust the amount of light provided to a building space. Security subsystem 438 can include occupancy sensors, video surveillance cameras, digital video recorders, video processing servers, intrusion detection devices, access control devices and servers, or other security-related devices.

Still referring to FIG. 4, BAS controller 366 is shown to include a communications interface 407 and a BAS interface 409. Interface 407 can facilitate communications between BAS controller 366 and external applications (e.g., monitoring and reporting applications 422, enterprise control applications 426, remote systems and applications 444, applications residing on client devices 448, etc.) for allowing user control, monitoring, and adjustment to BAS controller 366 and/or subsystems 428. Interface 407 can also facilitate communications between BAS controller 366 and client devices 448. BAS interface 409 can facilitate communications between BAS controller 366 and building subsystems 428 (e.g., HVAC, lighting security, lifts, power distribution, business, etc.).

Interfaces 407, 409 can be or include wired or wireless communications interfaces (e.g., jacks, antennas, transmitters, receivers, transceivers, wire terminals, etc.) for conducting data communications with building subsystems 428 or other external systems or devices. In various embodiments, communications via interfaces 407, 409 can be direct (e.g., local wired or wireless communications) or via a communications network 446 (e.g., a WAN, the Internet, a cellular network, etc.). For example, interfaces 407, 409 can include an Ethernet card and port for sending and receiving

data via an Ethernet-based communications link or network. In another example, interfaces **407**, **409** can include a Wi-Fi transceiver for communicating via a wireless communications network. In another example, one or both of interfaces **407**, **409** can include cellular or mobile phone communications transceivers. In one embodiment, communications interface **407** is a power line communications interface and BAS interface **409** is an Ethernet interface. In other embodiments, both communications interface **407** and BAS interface **409** are Ethernet interfaces or are the same Ethernet interface.

Still referring to FIG. 4, BAS controller **366** is shown to include a processing circuit **404** including a processor **406** and memory **408**. Processing circuit **404** can be communicably connected to BAS interface **409** and/or communications interface **407** such that processing circuit **404** and the various components thereof can send and receive data via interfaces **407**, **409**. Processor **406** can be implemented as a general purpose processor, an application specific integrated circuit (ASIC), one or more field programmable gate arrays (FPGAs), a group of processing components, or other suitable electronic processing components.

Memory **408** (e.g., memory, memory unit, storage device, etc.) can include one or more devices (e.g., RAM, ROM, Flash memory, hard disk storage, etc.) for storing data and/or computer code for completing or facilitating the various processes, layers and modules described in the present application. Memory **408** can be or include volatile memory or non-volatile memory. Memory **408** can include database components, object code components, script components, or any other type of information structure for supporting the various activities and information structures described in the present application. According to an exemplary embodiment, memory **408** is communicably connected to processor **406** via processing circuit **404** and includes computer code for executing (e.g., by processing circuit **404** and/or processor **406**) one or more processes described herein.

In some embodiments, BAS controller **366** is implemented within a single computer (e.g., one server, one housing, etc.). In various other embodiments BAS controller **366** can be distributed across multiple servers or computers (e.g., that can exist in distributed locations). Further, while FIG. 4 shows applications **422** and **426** as existing outside of BAS controller **366**, in some embodiments, applications **422** and **426** can be hosted within BAS controller **366** (e.g., within memory **408**).

Still referring to FIG. 4, memory **408** is shown to include an enterprise integration layer **410**, an automated measurement and validation (AM&V) layer **412**, a demand response (DR) layer **414**, a fault detection and diagnostics (FDD) layer **416**, an integrated control layer **418**, and a building subsystem integration layer **420**. Layers **410-420** can be configured to receive inputs from building subsystems **428** and other data sources, determine optimal control actions for building subsystems **428** based on the inputs, generate control signals based on the optimal control actions, and provide the generated control signals to building subsystems **428**. The following paragraphs describe some of the general functions performed by each of layers **410-420** in BAS **400**.

Enterprise integration layer **410** can be configured to serve clients or local applications with information and services to support a variety of enterprise-level applications. For example, enterprise control applications **426** can be configured to provide subsystem-spanning control to a graphical user interface (GUI) or to any number of enterprise-level business applications (e.g., accounting systems, user identification systems, etc.). Enterprise control appli-

cations **426** can also or alternatively be configured to provide configuration GUIs for configuring BAS controller **366**. In yet other embodiments, enterprise control applications **426** can work with layers **410-420** to optimize building performance (e.g., efficiency, energy use, comfort, or safety) based on inputs received at interface **407** and/or BAS interface **409**.

Building subsystem integration layer **420** can be configured to manage communications between BAS controller **366** and building subsystems **428**. For example, building subsystem integration layer **420** can receive sensor data and input signals from building subsystems **428** and provide output data and control signals to building subsystems **428**. Building subsystem integration layer **420** can also be configured to manage communications between building subsystems **428**. Building subsystem integration layer **420** translate communications (e.g., sensor data, input signals, output signals, etc.) across a plurality of multi-vendor/multi-protocol systems.

Demand response layer **414** can be configured to optimize resource usage (e.g., electricity use, natural gas use, water use, etc.) and/or the monetary cost of such resource usage in response to satisfy the demand of building **10**. The optimization can be based on time-of-use prices, curtailment signals, energy availability, or other data received from utility providers, distributed energy generation systems **424**, from energy storage **427** (e.g., hot TES **242**, cold TES **244**, etc.), or from other sources. Demand response layer **414** can receive inputs from other layers of BAS controller **366** (e.g., building subsystem integration layer **420**, integrated control layer **418**, etc.). The inputs received from other layers can include environmental or sensor inputs such as temperature, carbon dioxide levels, relative humidity levels, air quality sensor outputs, occupancy sensor outputs, room schedules, and the like. The inputs can also include inputs such as electrical use (e.g., expressed in kWh), thermal load measurements, pricing information, projected pricing, smoothed pricing, curtailment signals from utilities, and the like.

According to an exemplary embodiment, demand response layer **414** includes control logic for responding to the data and signals it receives. These responses can include communicating with the control algorithms in integrated control layer **418**, changing control strategies, changing setpoints, or activating/deactivating building equipment or subsystems in a controlled manner. Demand response layer **414** can also include control logic configured to determine when to utilize stored energy. For example, demand response layer **414** can determine to begin using energy from energy storage **427** just prior to the beginning of a peak use hour.

In some embodiments, demand response layer **414** includes a control module configured to actively initiate control actions (e.g., automatically changing setpoints) which minimize energy costs based on one or more inputs representative of or based on demand (e.g., price, a curtailment signal, a demand level, etc.). In some embodiments, demand response layer **414** uses equipment models to determine an optimal set of control actions. The equipment models can include, for example, thermodynamic models describing the inputs, outputs, and/or functions performed by various sets of building equipment. Equipment models can represent collections of building equipment (e.g., subplants, chiller arrays, etc.) or individual devices (e.g., individual chillers, heaters, pumps, etc.).

Demand response layer **414** can further include or draw upon one or more demand response policy definitions (e.g., databases, XML files, etc.). The policy definitions can be

edited or adjusted by a user (e.g., via a graphical user interface) so that the control actions initiated in response to demand inputs can be tailored for the user's application, desired comfort level, particular building equipment, or based on other concerns. For example, the demand response policy definitions can specify which equipment can be turned on or off in response to particular demand inputs, how long a system or piece of equipment should be turned off, what setpoints can be changed, what the allowable set point adjustment range is, how long to hold a high demand setpoint before returning to a normally scheduled setpoint, how close to approach capacity limits, which equipment modes to utilize, the energy transfer rates (e.g., the maximum rate, an alarm rate, other rate boundary information, etc.) into and out of energy storage devices (e.g., thermal storage tanks, battery banks, etc.), and when to dispatch on-site generation of energy (e.g., via fuel cells, a motor generator set, etc.).

Integrated control layer **418** can be configured to use the data input or output of building subsystem integration layer **420** and/or demand response layer **414** to make control decisions. Due to the subsystem integration provided by building subsystem integration layer **420**, integrated control layer **418** can integrate control activities of the subsystems **428** such that the subsystems **428** behave as a single integrated supersystem. In an exemplary embodiment, integrated control layer **418** includes control logic that uses inputs and outputs from a plurality of building subsystems to provide greater comfort and energy savings relative to the comfort and energy savings that separate subsystems could provide alone. For example, integrated control layer **418** can be configured to use an input from a first subsystem to make an energy-saving control decision for a second subsystem. Results of these decisions can be communicated back to building subsystem integration layer **420**.

Integrated control layer **418** is shown to be logically below demand response layer **414**. Integrated control layer **418** can be configured to enhance the effectiveness of demand response layer **414** by enabling building subsystems **428** and their respective control loops to be controlled in coordination with demand response layer **414**. This configuration can reduce disruptive demand response behavior relative to conventional systems. For example, integrated control layer **418** can be configured to assure that a demand response-driven upward adjustment to the setpoint for chilled water temperature (or another component that directly or indirectly affects temperature) does not result in an increase in fan energy (or other energy used to cool a space) that would result in greater total building energy use than was saved at the chiller.

Integrated control layer **418** can be configured to provide feedback to demand response layer **414** so that demand response layer **414** checks that constraints (e.g., temperature, lighting levels, etc.) are properly maintained even while demanded load shedding is in progress. The constraints can also include setpoint or sensed boundaries relating to safety, equipment operating limits and performance, comfort, fire codes, electrical codes, energy codes, and the like. Integrated control layer **418** is also logically below fault detection and diagnostics layer **416** and automated measurement and validation layer **412**. Integrated control layer **418** can be configured to provide calculated inputs (e.g., aggregations) to these higher levels based on outputs from more than one building subsystem.

Automated measurement and validation (AM&V) layer **412** can be configured to verify that control strategies commanded by integrated control layer **418** or demand

response layer **414** are working properly (e.g., using data aggregated by AM&V layer **412**, integrated control layer **418**, building subsystem integration layer **420**, FDD layer **416**, or otherwise). The calculations made by AM&V layer **412** can be based on building system energy models and/or equipment models for individual BAS devices or subsystems. For example, AM&V layer **412** can compare a model-predicted output with an actual output from building subsystems **428** to determine an accuracy of the model.

Fault detection and diagnostics (FDD) layer **416** can be configured to provide on-going fault detection for building subsystems **428**, building subsystem devices (i.e., building equipment), and control algorithms used by demand response layer **414** and integrated control layer **418**. FDD layer **416** can receive data inputs from integrated control layer **418**, directly from one or more building subsystems or devices, or from another data source. FDD layer **416** can automatically diagnose and respond to detected faults. The responses to detected or diagnosed faults can include providing an alarm message to a user, a maintenance scheduling system, or a control algorithm configured to attempt to repair the fault or to work-around the fault.

FDD layer **416** can be configured to output a specific identification of the faulty component or cause of the fault (e.g., loose damper linkage) using detailed subsystem inputs available at building subsystem integration layer **420**. In other exemplary embodiments, FDD layer **416** is configured to provide "fault" events to integrated control layer **418** which executes control strategies and policies in response to the received fault events. According to an exemplary embodiment, FDD layer **416** (or a policy executed by an integrated control engine or business rules engine) can shut-down systems or direct control activities around faulty devices or systems to reduce energy waste, extend equipment life, or assure proper control response.

FDD layer **416** can be configured to store or access a variety of different system data stores (or data points for live data). FDD layer **416** can use some content of the data stores to identify faults at the equipment level (e.g., specific chiller, specific AHU, specific terminal unit, etc.) and other content to identify faults at component or subsystem levels. For example, building subsystems **428** can generate temporal (i.e., time-series) data indicating the performance of BAS **400** and the various components thereof. The data generated by building subsystems **428** can include measured or calculated values that exhibit statistical characteristics and provide information about how the corresponding system or process (e.g., a temperature control process, a flow control process, etc.) is performing in terms of error from its setpoint. These processes can be examined by FDD layer **416** to expose when the system begins to degrade in performance and alarm a user to repair the fault before it becomes more severe.

Autonomous FDD Rule Threshold Tuning

Referring now to FIG. 5, a block diagram of a system **500** including a false positive and false negative prediction system **502** that is configured to predict false positives and false negatives of faults detected in building equipment by the FDD layer **416** in a building management system (e.g., BMS **400**) is shown, according to an exemplary embodiment. False positive and false negative prediction system **502** may operate in a cloud environment or locally by a processor at the building management system. False positive and false negative prediction system **502** may implement one or more machine learning models to predict false positives and false negatives of faults that are detected in the building equipment. False positive and false negative pre-

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diction system **502** may do so by inputting measurements of various points of the piece of building equipment into the machine learning models and determining whether individual output confidence scores for false positives and false negatives from the models satisfy a predetermined criteria (e.g., exceed a predetermined threshold, is the highest predicted confidence score, etc.).

According to various example implementations of the present disclosure, the FDD layer **416** may determine that a building equipment has a fault based on the trigger or satisfaction of a rule. A false positive may indicate that the building equipment actually is not experiencing a fault, and the rule incorrectly determined that a fault has occurred. Furthermore, the FDD layer **416** may determine that a building equipment is not experiencing a fault because a state of the building did not satisfy the rule. A false negative may indicate that the building equipment is actually experiencing a fault, and the rule incorrectly determined that the fault did not occur. Although the false positive prediction model **528** is described in detail herein, similar description may apply for the false negative prediction model **530** except for false negatives instead of false positives.

As used herein, “points” or “data points” refer to sensor inputs, control outputs, control values, and/or different characteristics of the inputs and/or outputs. “Points” and/or “data points” may refer to various data objects relating to the inputs and the outputs such as BACnet objects. The objects may represent and/or include a point and/or group of points. The object may include various properties for each of the points. For example, an analog input may be a particular point represented by an object with one or more properties describing the analog input and another property describing the sampling rate of the analog input. For example, in some embodiments, a point is a data representation associated with a component of a BMS, such as a camera, thermostat, controller, VAV box, RTU, valve, damper, chiller, boiler, AHU, supply fan, etc.

System **500** may include a user presentation system **538**, a building controller **540**, and building equipment **542**. Building controller **540** may be similar to or the same as BMS controller **366**. False positive and false negative prediction system **502** may be a component of or be within building controller **540**. In some embodiments, false positive and false negative prediction system **502** operates in the cloud as one or more cloud servers. Components **502** and **538-542** may communicate over a network (e.g., a synchronous or asynchronous network).

False positive and false negative prediction system **502** may include a processing circuit **504**, a processor **506**, and a memory **508**. Processing circuit **504**, processor **506**, and/or memory **508** can be the same as, or similar to, processing circuit **404**, processor **406**, and/or memory **408**, as described with reference to FIG. 4. Memory **508** may include a data pre-processor **510**, equipment models **512a-n**, a training manager **514**, a data post-processor **516**, and a measurement database **518**. Memory **508** may include any number of components.

Data pre-processor **510** includes instructions performed by one or more servers or processors (e.g., processing circuit **504**), in some embodiments. In some embodiments, data pre-processor **510** includes a data collector **522**, a vector generator **524**, and an FP/FN identifier **526**. Data collector **522** may be configured to collect data that corresponds to different pieces of building equipment (e.g., building equipment **542**). Data collector **522** can be configured to retrieve and/or collect building data from a building management system and store the building data in measurement database

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518, in some embodiments. Data collector **522** can be configured to collect data automatically or, in some embodiments, poll sensors associated with building equipment **542** to collect data at predetermined time intervals set by an administrator. In some embodiments, data collector **522** can further be configured to collect data upon detecting that a value changed by an amount exceeding a threshold. In some embodiments, data collector **522** is configured to collect building data upon receiving a request from an administrator. The administrator may make the request from a client device. The administrator can request building data associated with any time period and building device.

In some embodiments, the data collector **522** may execute rules related to the building equipment and determine whether any rules have been satisfied. For example, a rule pertaining to the AHU may state that if the air supply temperature is less than a first threshold and if the fan speed is greater than a second threshold, the AHU has a fault. Accordingly, the data collector **522** may collect the measurements (e.g., air supply temperature and fan speed) from the measurement database **518** and determine whether any rule is satisfied such that a fault is determined to exist for the AHU. Then, the data collector **522** may provide the measurements, the thresholds, and the determination of whether there is a fault to the vector generator **524** so that a vector may be generated for the prediction models.

Data collector **522** may be configured to tag each data point of the data with timestamps indicating when the data point was generated and/or when data collector **522** collected the data point from the sensors. In some embodiments, data collector **522** can also tag the data with a device identifier tag indicating the building device from which the building data was collected. Thus, data collector **522** may store the timestamped data in measurement database **518** as a timeseries corresponding to how the measured values changed over time.

As described herein, timeseries can be a collection of values for a particular point (e.g., a discharge air temperature point of an air handling unit, a discharge air temperature, a supply fan status, a zone air temperature, a humidity, a pressure, etc.) generated at different times (e.g., at periodic intervals). The values may include or be associated with identifiers of the building devices with which the points are associated (e.g., an air handler, a VAV box, a controller, a chiller, a boiler, vents, dampers, etc.). Each timeseries can include a series of values for the same point and a timestamp for each of the data values. For example, a timeseries for a point provided by a temperature sensor (e.g., provided through local gateways) can include a series of temperature values measured by the temperature sensor and the corresponding times at which the temperature values were measured. An example of a timeseries which can be generated by data collector **522** is as follows:

```
[<key, timestamp1, value1>, <key, timestamp2, value2>,
<key, timestamp3, value3>]
```

where key is an identifier of the source of the raw data samples (e.g., timeseries ID, sensor ID, device ID, etc.), timestamp_i may identify the time at which the ith sample was collected, and value_i may indicate the value of the ith sample.

Measurement database **518** may be a database configured to store building data associated with a building management system (e.g., BMS **400**). Measurement database **518** can be a graph database, MySQL, Oracle, Microsoft SQL, PostgreSQL, DB2, document store, search engine, device identifier-value store, etc. Measurement database **518** can be configured to hold data including any amount of values and

can be made up of any number of components. The data can include various measurements and states (e.g., temperature readings, pressure readings, device state readings, blade speeds, etc.) associated with building equipment (e.g., AHUs, chillers, boilers, VAVs, fans, etc.) of the building management system. In some embodiments, the building data is tagged with timestamps indicating times and dates that the values of the building data were generated by devices (e.g., sensors) of the building management system or retrieved by data collector 522. It should be understood that, in some embodiments, measurement database 518 or any other type of data or data structure described herein may be or include a data structure configured to store digital twins of the building, building equipment, building spaces, building occupants/people, events, or any other entity of or related to the building. For example, measurement database 518 and/or other elements described herein may be implemented as a knowledge graph having nodes representing entities of the building and edges representing relationships between the entities. Example implementations of such digital twins and knowledge graphs as may be utilized in conjunction with the features of the present disclosure can be found in U.S. patent application Ser. No. 17/529,118, filed Nov. 18, 2021, which is incorporated herein by reference in its entirety.

In some embodiments, measurement database 518 may store setpoint values for different points of the building management system. The stored setpoint values may be associated with a schedule indicating the times in which building equipment 542 will operate so points of the building management system will reach the corresponding stored setpoints. For example, a setpoint schedule may indicate that a kitchen should be 70 degrees at 7 P.M. but 68 degrees at 3 P.M. Accordingly, a controller (e.g., building controller 540) may control the building equipment of the building to cause the temperature point to reach the setpoint temperature at the corresponding times. Measurement database 518 may include schedules for setpoints of any point of the building to reach a desired level of comfort for the building's occupants.

Vector generator 524 may be configured to generate a feature vector that is configured to be input into machine learning models of equipment models 512a-n from measurement database 518. For example, the feature vector may include a state of the building equipment and a threshold that is used in the rule. For example, for an AHU of the building, there may be a rule that states if supply air temperature is less than a certain temperature and the fan speed is greater than a certain rpm, there is a fault with the AHU. This may rule may be expressed as follows:

if ($c_1 < \epsilon_1$ & $c_2 > \epsilon_2$),

then AHU has fault F_1

where c_1 corresponds to the supply air temperature, ϵ_1 corresponds to a first threshold, c_2 corresponds to the fan speed, ϵ_2 corresponds to a second threshold, and F_1 corresponds to the fault ID. For a first machine learning model, the state of the AHU may include the supply air temperature and the first threshold, and for a second machine learning model, the state of the AHU may include the fan speed and the second threshold. These pairs may be provided as feature vectors to the respective machine learning models.

Vector generator 524 may generate such feature vectors upon determining an event has occurred. An event may be or include a detection that a value associated with the piece of building equipment is above a threshold, a determination that a predetermined time interval has passed since vector generator 524 previously executed the machine learning

model, receipt of a user input indicating to execute the machine learning model, receipt of a signal from another computing device indicating to execute the machine learning model, etc. Vector generator 524 may monitor various aspects of the building management system to identify such events and determine when the events occur. For example, vector generator 524 may keep track of the times in which vector generator 524 executes the machine learning model. Vector generator 524 may maintain an internal clock and identify when a predetermined (e.g., a pre-programmed) time period has passed since the last time vector generator 524 executed the machine learning model and determine the predetermined time period has passed. Vector generator 524 may identify an event as occurring upon determining the predetermined time period has passed.

Upon determining an event has occurred, vector generator 524 may generate a feature vector. Vector generator 524 may generate the feature vector by identifying the piece of building equipment that is associated with the event (e.g., the piece of building equipment that has a stored association with the event) and retrieve data that corresponds to the piece of building equipment. Vector generator 524 may retrieve the data that is associated with attributes or points of the piece of building equipment based on a stored association between the values and the attributes or points. Vector generator 524 may retrieve data that is associated with values from within a pre-configured time frame of the event (e.g., values that are associated with timestamps from a time frame before and/or after the event) and generate a feature vector using the retrieved values. Vector generator 524 may retrieve values that were collected from sensors of the building and/or values of setpoints that are stored in memory (e.g., measurement database 518).

Upon generating the feature vector, vector generator 524 may identify the machine learning model that is associated with the piece of building equipment that is associated with the event. Vector generator 524 may identify the machine learning model from equipment models 512a-n that each includes or is otherwise associated with a different false positive prediction model 528 and/or a false negative prediction model 530. Each of equipment models 512a-n may be a data representation of a different piece of building equipment within the building management system. The false positive prediction models and/or false negative prediction models of each equipment model 512a-n may be associated with a device identifier of the respective equipment model 512a-n. Vector generator 524 may identify, responsive to determining the identified event, false positive prediction model 528 and false negative prediction model 530 are associated with the same or an identical device identifier. Upon identifying false positive prediction model 528, vector generator 524 may apply the generated feature vector to false positive prediction model 528 and execute false positive prediction model 528.

False positive prediction model 528 may be a machine learning model (e.g., a neural network, a random forest, a support vector machine, etc.) configured to output confidence scores associated with a whether the satisfaction of a fault rule (e.g., determination that the building equipment has a fault) related to a building equipment was a false positive. False positive prediction model 528 may be configured to output confidence scores for fault determinations based on feature vectors that are generated by vector generator 524 based on data that corresponds to a particular piece of building equipment (e.g., the piece of building equipment that the equipment model represents). False positive prediction model 528 may output confidence scores

for whether the determination that a fault has occurred in the building equipment is a false positive. FP/FN identifier **526** may identify the confidence scores and/or determine whether the determination of a fault in the piece of building equipment is a false positive in the future based on the confidence scores.

FP/FN identifier **526** may be configured to use a predetermined criteria to determine if and/or when a fault is likely to occur in a piece of building equipment. The predetermined criteria may be a threshold and/or one or more rules. For instance, FP/FN identifier **526** may determine a fault determination is a false positive by comparing the confidence score to a predetermined threshold. Responsive to determining the score exceeds the threshold, FP/FN identifier **526** may determine that the fault determination (e.g., the fact that there was a fault in the building equipment) is a false positive. However, responsive to determining the score does not exceed the threshold, the data processing system may determine that the fault determination is not likely a false positive (e.g., the fault is actually a true positive). The data processing system may compare the confidence score to any rule or threshold.

Upon determining a confidence score for a rule satisfied the predetermined criteria, FP/FN identifier **526** may identify the rule that was determined to have been satisfied and associated with the confidence score and an identification of the threshold in the rule that was used to determine the fault. In some embodiments, FP/FN identifier **526** may generate an alert indicating the fault did not occur in the building equipment and transmit the alert to a client device (e.g., an administrative device) so an administrator can view the alert and keep the building equipment operation/online. In some embodiments, FP/FN identifier **526** may feed the identification of the false positive back to vector generator **524**, which in turn can use the identification to generate a new feature vector.

False negative prediction model **530** may be a machine learning model similar to false positive prediction model **528** that is configured to predict when false negatives occurred. False negative prediction model **530** may be configured to output confidence scores for fault determinations based on feature vectors that are generated by vector generator **524** based on data that corresponds to a particular piece of building equipment (e.g., the piece of building equipment that the equipment model represents). False positive prediction model **528** may output confidence scores for false positives and false negatives indicating likelihoods that the individual false positives and false negatives are the correct prediction.

Threshold optimizer **542** may receive the output confidence scores and process the scores to optimize the threshold that was used to determine the faults in the rules that were used in the false positive prediction model **528** and the false negative prediction model **530**. The threshold optimizer **542** may use constrained nonlinear optimization to find the optimal threshold value (based on the accumulated data so far) that minimizes the number of false positives and false negatives.

Data post-processor **516** may receive the output confidence scores and the optimal threshold values and process the scores and thresholds to transmit a signal to user presentation system **538** and/or building controller **540** to adjust the threshold of the rules. Data post-processor **516** includes instructions performed by one or more servers or processors (e.g., processing circuit **504**), in some embodiments. In some embodiments, data post-processor **516** includes a record generator **536**. Record generator **536** may receive the

predicted confidence scores and/or optimal threshold value and generate a record (e.g., a file, document, table, listing, message, notification, etc.) including confidence scores and/or the threshold values. Upon generating the record, record generator **536** may transmit the record to user presentation system **538** for display and/or building controller **540** to use to adjust operation or the configuration of building equipment **542** to avoid the false positive and false negative in the rule.

False positive and false negative prediction system **502** can provide indications of recommendations of how to adjust the threshold values to user presentation system **538** and/or building controller **540**. In some embodiments, building controller **540** uses the expected recommendations to operate building equipment **542** (e.g., control environmental conditions of a building, cause generators to turn on or off, charge or discharge batteries, etc.). Further, user presentation system **538** can receive the indications and/or recommendations and cause a client device to display indications (e.g., graphical elements, charts, words, numbers, etc.) of the threshold values and/or recommendations. For example, user presentation system **538** may receive the rule and the optimal threshold value so that the user can modify the rule in the FDD layer **416**.

In some embodiments, false positive and false negative prediction system **502** trains the prediction models of equipment models **512a-n** using training manager **514**. Training manager **514** includes instructions performed by one or more servers or processors (e.g., processing circuit **504**), in some embodiments. In some embodiments, training manager **514** includes a false positive prediction model trainer **532** and/or a false negative prediction model trainer **534**. False positive prediction model trainer **532** may be configured to train false positive prediction model **528** and other false positive prediction models of equipment models **512a-n** to predict when false positives occur when a fault is detected for pieces of building equipment. False positive prediction model trainer **532** may feed labeled training data including measurements associated with points of a particular piece of building equipment to the false positive prediction model associated with the piece of building equipment. The respective false positive prediction model may output confidence scores for thresholds and false positive prediction model trainer **532** may determine differences between the predicted outputs and the labels and use back-propagation techniques according to a loss function to adjust the false positive prediction model's weights and parameters proportional to the determined differences. False positive prediction model trainer **532** may repeat these steps for any number of fault prediction machine learning models to train the machine learning models to predict future false positives for individual pieces of building equipment.

Similarly, false negative prediction model trainer **534** may be configured to train false negative prediction model **530** and other false negative prediction models of equipment models **512a-n**. False negative prediction model trainer **534** may feed measurement data and/or identifications of false negatives to obtain confidence scores for the threshold in rules used in determining faults in the building equipment. False negative prediction model trainer **534** may identify labels indicating the correct output, determine differences between the correct output and the respective false negative prediction model's output, and use back-propagation techniques according to a loss function to adjust the false negative prediction model's weights and parameters according to the determined differences. False negative prediction model trainer **534** may repeat these steps for any number of

false negative prediction models to the machine learning models to predict false negatives for individual pieces of building equipment. In some embodiments, false negative prediction model trainer **534** may train a false negative prediction model in real-time. In such embodiments, false negative prediction model trainer **534** may feed measurement data and/or identifications of false negatives into a false negative prediction model to obtain confidence scores for false negatives in a rule.

A user may input levels of accuracy (e.g., correct, incorrect, partially correct, etc.) of the recommendations and/or the predicted false positives or false negatives. False positive prediction model trainer **532** or false negative prediction model trainer **534** may identify the input levels of accuracy, determine differences between the predicted confidence scores and the input levels of accuracy, and use back-propagation techniques with the false negative prediction model that predicted the confidence scores for the false positives and false negatives according to a loss function based on the differences. Thus, false negative prediction model trainer **534** may train false negative prediction models in real-time or near real-time, in some embodiments.

In some embodiments, training manager **514** may operate in a cloud server and be configured to use training data from multiple building management systems to train false positive prediction models and/or false negative prediction models. Training manager **514** may be configured to train individual machine learning models using training data that is associated with multiple pieces of building equipment (e.g., building equipment of the same type) until the machine learning models are accurate to a threshold, and then deploy the machine learning models to the local building management system to be used to make predictions for individual pieces of building equipment (and be further trained based only on data associated with the piece of building equipment). This may be advantageous in building management systems that do not have enough training data to train machine learning models to make accurate predictions.

In such embodiments, training manager **514** may be configured to train the machine learning models using a weighting policy. The weight policy may include weights that can be applied to different training data sets. The weights may correspond to different building management systems and may be determined based on how trustworthy an administrator has determined data from a building management system to be and/or based on whether the data originated at a building management system for which the models are being trained. Training manager **514** may use the weights by weighting the differences in a loss function so that training data that is associated with higher weights cause higher shifts in the weights or parameters of a machine learning model than training data that is associated with lower weights during training. Thus, training manager **514** may control the training to improve the accuracy and speed with which machine learning models are trained to be employed at individual building management systems.

Referring now to FIG. 6, a system **600** of a rule (or trigger rule) **602** of a building equipment where parameters of the trigger rule **602** are trained is shown, according to an exemplary embodiment. In some embodiments, the system **600** is similar to the system **400** or system **500**. In the example described in FIG. 6 below, a simple rule for an AHU is described for the sake of clarity and brevity, but embodiments are not limited thereto, and the disclosed technology may be implemented for a variety of other building equipment and/or rules. For example, the building equipment may be related to fire safety **430**, lifts/escalators

432, electrical **434**, ICT **436**, security **438**, HVAC **440**, lighting **442** or any other building equipment within or around buildings. Further, the rules may be related to any rule that is implemented within the FDD layer **416**.

The system **600** can perturb parameters, ϵ_1 of the trigger rule **602**. For example, the trigger rule **602** may include a rule that if supply air temperature is less than ϵ_1 °C., the rule is triggered (or satisfied), and a corresponding action be performed. The corresponding action can be a determination that a fault exists with the AHU. The perturbation of the parameters can be increasing or decreasing the parameters in set amounts from existing values. The perturbation of the parameters can be selecting a space of values for the parameters and/or randomizing the parameters and/or parameter space.

With the perturbed values for ϵ_1 , the BMS controller **366** can simulate the state of the AHU for various temperatures within a range. The simulation can be performed by the false positive and false negative prediction system **502** via the models **512a-512n**. The output of the models **512a-512n** can be false positive or false negative.

The system **600** can analyze the states produced by the building equipment operation **604** to determine states of the building equipment. For example, the air supply temperature and fan speed can be generated and collected for each state. Furthermore, the state can be any metric that is collected from any portion of the building equipment. For example, the state can include supply air temperature, fan speed, AHU operating mode (heating, cooling with outdoor air, etc.), outside air temperature, zone temperature, zone humidity, etc. Once the state is collected, various rules may be calculated related to the building equipment to determine whether a fault exists.

The system **600** can receive feedback **606** from a user on whether the fault that was computed is a false positive, false negative, or true positive. As discussed above, false positive can imply that although a rule was satisfied to determine that a fault existed, there is no fault. False negative can imply that although the rule was not satisfied, and therefore no fault was detected, the building equipment in fact does have a fault. And a true positive can imply that the fault that was detected is true and no adjustment needs to be made to the rule.

The system **600** can generate accumulated training data **608**. The accumulated training data can include the values of the parameters ϵ_1 , the state of the building equipment for each value of the parameters, and the labeled data including whether the fault determination was a false positive, false negative, or true positive.

The system **600** can generate neural networks **610** for predicting whether the fault determination was a false positive or a false negative based on the state s and parameters ϵ . The neural networks **610** can be trained by the system **600** based on the accumulated training data **608**.

The system **600** can then determine optimal values for the parameters ϵ . The system **600** can search a space of potential values for ϵ that consider false positives and false negatives predicted by the trained neural network models **610**. The optimization can be the relation **800** shown in FIG. 8. The optimization **612** performed by the system **600** can be a method of computing the optimal threshold of a trigger condition using the neural network models **610** of rewards and solving a constrained nonlinear optimization model. In some embodiments, the optimal values for the parameters found by the system **600** can be presented to a user for review and/or approval via a user interface, e.g., via the client device **368** or **448**.

Referring now to FIG. 7, a process 700 for identifying values for the parameters of the trigger rule 602 of FIG. 6 is shown, according to an exemplary embodiment. The process 700 can be performed by the system 600 and/or any component of the system 600. Furthermore, the process 700 can be performed by any computing device described herein.

In step 702, the system 600 can perturb a building equipment with various values for thresholds ϵ . The result of the perturbed parameters can result in various states, s . The states can be states of the building equipment as described herein. The perturbations can result in pairs (s, ϵ) that can be used to determine feedback for false positive, false negative, or true positive.

In step 704, the system 600 can accumulate the pairs and corresponding feedback to create neural network models, e.g., the neural networks 610 based on the data determined in step 702. The neural networks 610 can predict false positive and false negative rewards as a function of the state and the threshold, e.g., $FP=f(s, \epsilon)$ and $FN=f(s, \epsilon)$.

In step 706, the system 600 can determine a value for the parameter, ϵ that minimizes a relation, $(\alpha_1 \cdot FP + \alpha_2 \cdot FN)$. The minimization is shown in relation 800 of FIG. 8. The values of α_1 and α_2 can weigh the various rewards in the relation that is minimized, e.g., the false positive and false negative reward.

In step 708, the system 600 can periodically repeat the steps 702-706. For example, the system 600 can repeat the steps at a defined time period (e.g., every day, week, month, etc.). In some embodiments, the retraining may occur if the rate of false positives and/or the rate of false negatives exceed an acceptable range set by the user. For example, if the user set the acceptable false positive rate to be 10% or lower, and the neural network model for false positives had a rate of 15%, the model may be retrained.

Referring now to FIG. 9, a process 900 of determining the optimal threshold for a rule (e.g., building equipment rule, trigger rule, or fault rule) is shown, according to an exemplary embodiment. The process 900 can be performed by the system 400, 500, or 600 and/or any of their components or a combination of the components. The process 900 can be performed by any computing device or processing circuit described herein. Furthermore, the process 900 may be include additional steps or certain steps may be removed depending on embodiments.

In step 902, the system can provide a rule including a threshold. The rule can pertain to any that is used by the FDD layer 416 to determine whether a fault exists within the building equipment or not. For example, if there is a rule that states if AHU mode is mechanical cooling with 100% outdoor air and the magnitude of the difference between outside air temperature and mixed air temperature is greater than a threshold, the AHU is faulty.

In step 904, the system can receive a state of the building equipment. In some embodiments, the state can include any or all values that are used to calculate whether a rule is satisfied or not. In the above example, the AHU running in a mode with mechanical cooling with 100% outdoor air, the outside air temperature, and mixed air temperature can all be part of the state of the AHU. Further, the threshold can be the value that is being optimized so that the system can reduce or minimize the number of false positives and false negatives.

In step 906, the system can assess, using a machine learning model, whether the determination of whether the building equipment has fault is a false positive or a false negative based on the state and the threshold. For example, if the above rule was satisfied, the system may determine

that a fault exists in the AHU. However, in reality, the AHU may be operating normally and not have a fault as determined by occupants of the building or the building manager. Accordingly, the model's administrator may provide feedback to the system indicating that the fault is a false positive. In another example, the rule may not be satisfied and the system may then determine that the AHU does not have a fault. However, based on occupant or building manager feedback, it may be determined that the AHU is actually faulty, and the administrator may provide feedback to the system that there was a false negative.

In step 908, the system may determine a new threshold based on the assessment of the machine learning model. This may be done using constrained nonlinear optimization that solves for the minimal number of false positives and false negatives based on the feedback received in step 906. The new threshold may be higher or lower than the existing one, depending on the parameters set with the optimization equations. Further, the user may desire to reduce the number of false positives more than the false negatives, in which case the optimization parameters used in the equations may be adjust so that greater weight is given to minimizing the false positives. In other embodiments, the user may be interested in reducing the false negatives more than the false positives, in which case greater weight can be given to reducing the false negatives more than the false positives.

In step 910, the system may replace the threshold with the new threshold to make subsequent determinations of whether the building equipment or other building equipment has a fault. The number of false positives and false negatives may be reduced and/or minimized.

Configuration of Exemplary Embodiments

The construction and arrangement of the systems and methods as shown in the various exemplary embodiments are illustrative only. Although only a few embodiments have been described in detail in this disclosure, many modifications are possible (e.g., variations in sizes, dimensions, structures, shapes and proportions of the various elements, values of parameters, mounting arrangements, use of materials, colors, orientations, etc.). For example, the position of elements may be reversed or otherwise varied and the nature or number of discrete elements or positions may be altered or varied. Accordingly, all such modifications are intended to be included within the scope of the present disclosure. The order or sequence of any process or method steps may be varied or re-sequenced according to alternative embodiments. Other substitutions, modifications, changes, and omissions may be made in the design, operating conditions and arrangement of the exemplary embodiments without departing from the scope of the present disclosure.

The present disclosure contemplates methods, systems and program products on any machine-readable media for accomplishing various operations. The embodiments of the present disclosure may be implemented using existing computer processors, or by a special purpose computer processor for an appropriate system, incorporated for this or another purpose, or by a hardwired system. Embodiments within the scope of the present disclosure include program products comprising machine-readable media for carrying or having machine-executable instructions or data structures stored thereon. Such machine-readable media can be any available media that can be accessed by a general purpose or special purpose computer or other machine with a processor. By way of example, such machine-readable media can comprise RAM, ROM, EPROM, EEPROM, CD-ROM or other optical disk storage, magnetic disk storage or other magnetic storage devices, or any other medium which can be used to

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carry or store desired program code in the form of machine-executable instructions or data structures and which can be accessed by a general purpose or special purpose computer or other machine with a processor. When information is transferred or provided over a network or another communications connection (either hardwired, wireless, or a combination of hardwired or wireless) to a machine, the machine properly views the connection as a machine-readable medium. Thus, any such connection is properly termed a machine-readable medium. Combinations of the above are also included within the scope of machine-readable media. Machine-executable instructions include, for example, instructions and data which cause a general purpose computer, special purpose computer, or special purpose processing machines to perform a certain function or group of functions.

Although the figures show a specific order of method steps, the order of the steps may differ from what is depicted. Also two or more steps may be performed concurrently or with partial concurrence. Such variation will depend on the software and hardware systems chosen and on designer choice. All such variations are within the scope of the disclosure. Likewise, software implementations could be accomplished with standard programming techniques with rule based logic and other logic to accomplish the various connection steps, processing steps, comparison steps and decision steps.

In various implementations, the steps and operations described herein may be performed on one processor or in a combination of two or more processors. For example, in some implementations, the various operations could be performed in a central server or set of central servers configured to receive data from one or more devices (e.g., edge computing devices/controllers) and perform the operations. In some implementations, the operations may be performed by one or more local controllers or computing devices (e.g., edge devices), such as controllers dedicated to and/or located within a particular building or portion of a building. In some implementations, the operations may be performed by a combination of one or more central or offsite computing devices/servers and one or more local controllers/computing devices. All such implementations are contemplated within the scope of the present disclosure. Further, unless otherwise indicated, when the present disclosure refers to one or more computer-readable storage media and/or one or more controllers, such computer-readable storage media and/or one or more controllers may be implemented as one or more central servers, one or more local controllers or computing devices (e.g., edge devices), any combination thereof, or any other combination of storage media and/or controllers regardless of the location of such devices.

What is claimed:

1. A non-transitory computer-readable storage medium having instructions stored thereon that, when executed by one or more processors, cause the one or more processors to: provide a rule including a threshold, wherein the rule is used to generate a determination of whether building equipment has a fault; receive a state of the building equipment; perturb the building equipment with multiple values of the threshold to provide additional data for a machine learning model; generate, by the machine learning model, an assessment of whether the determination of whether the building equipment has a fault is a false positive or a false negative based on the state and the threshold;

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determine a new threshold based on the assessment of the machine learning model; and replace the threshold with the new threshold to make subsequent determinations of whether the building equipment or other building equipment has a fault.

2. The non-transitory computer-readable storage medium of claim 1, wherein the new threshold reduces a number of false positives a number of false negatives.

3. The non-transitory computer-readable storage medium of claim 1, wherein the instructions further cause the one or more processors to accumulate training data for the machine learning model using the state and the threshold as inputs and labels of false positive or false negative as outputs.

4. The non-transitory computer-readable storage medium of claim 1, wherein the rule includes: a condition portion including a comparison of the state to the threshold; and an action portion including a determination that the building equipment is faulty when the condition is satisfied or not faulty when the condition is not satisfied.

5. The non-transitory computer-readable storage medium of claim 4, wherein the machine learning model includes a first machine learning model for predicting a false negative and a second machine learning model for predicting a false positive.

6. The non-transitory computer-readable storage medium of claim 5, wherein the first machine learning model is configured to accurately predict false positives at a first rate, and wherein the instructions further cause the one or more processors to retrain the first machine learning model when the first rate exceeds a first threshold rate.

7. The non-transitory computer-readable storage medium of claim 5, wherein the second machine learning model is configured accurately predict false negatives at a second rate, and wherein the instructions further cause the one or more processors to retrain the second machine learning model when the second rate exceeds a second threshold rate.

8. The non-transitory computer-readable storage medium of claim 1, wherein the one or more processors are configured to use a constrained nonlinear optimization to determine the new threshold.

9. A method comprising:

providing, by a processing circuit, a rule including a threshold, wherein the rule is used to generate a determination of whether building equipment has a fault; receiving, by the processing circuit, a state of the building equipment;

perturbing, by the processing circuit, the building equipment with multiple values of the threshold to provide additional data for a machine learning model;

generating, by the processing circuit and the machine learning model, an assessment of whether the determination of whether the building equipment has a fault is a false positive or a false negative based on the state and the threshold;

determining, by the processing circuit, a new threshold based on the assessment of the machine learning model; and

replacing, by the processing circuit, the threshold with the new threshold to make subsequent determinations of whether the building equipment or other building equipment has a fault.

10. The method of claim 9, wherein the new threshold reduces a number of false positives a number of false negatives.

11. The method of claim 9, wherein accumulating, by the processing circuit, training data for the machine learning

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model using the state and the threshold as inputs and labels of false positive or false negative as outputs.

12. The method of claim 9, wherein the rule includes:
a condition portion including a comparison of the state to the threshold; and

an action portion including a determination that the building equipment is faulty when the condition is satisfied or not faulty when the condition is not satisfied.

13. The method of claim 12, wherein the machine learning model includes a first machine learning model for predicting a false negative and a second machine learning model for predicting a false positive.

14. The method of claim 13, wherein the first machine learning model is configured to accurately predict false positives at a first rate, and wherein the method further comprises retraining, by the processing circuit, the first machine learning model when the first rate exceeds a first threshold rate.

15. The method of claim 13, wherein the second machine learning model is configured accurately predict false negatives at a second rate, and wherein the method further comprises retraining, by the processing circuit, the second machine learning model when the second rate exceeds a second threshold rate.

16. The method of claim 9, further comprising using, by the processing circuit, a constrained nonlinear optimization to determine the new threshold.

17. A building system comprising:

one or more storage devices storing instructions thereon; and

one or more processors, wherein the one or more processors execute the instructions causing the one or more processors to:

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provide a rule used to generate a determination of whether a building equipment has a fault;

perturb the building equipment having a plurality of states with a plurality of corresponding thresholds for the rule for determining whether a fault exists;

determine whether a fault exists based on the perturbed building equipment;

receive feedback of whether the determination of whether a fault exists is a false positive or a false negative, or a true positive;

provide training data to a machine learning model, wherein the training data includes the plurality of states and the plurality of corresponding thresholds as inputs and the feedback of false positive or false negative as outputs;

receive a current state of the building equipment and a current threshold of the rule;

generate, by the trained machine learning model, an assessment of whether the determination of whether the building equipment has a fault is a false positive or a false negative based on the current state and the current threshold;

determine a new threshold based on the assessment of the trained machine learning model; and

replace the current threshold with the new threshold to make subsequent determinations of whether the building equipment or other building equipment has a fault.

18. The building system of claim 17, wherein the instructions further cause the one or more processors to use constrained nonlinear optimization to find the new threshold that reduces a number of false positives a number of false negatives.

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